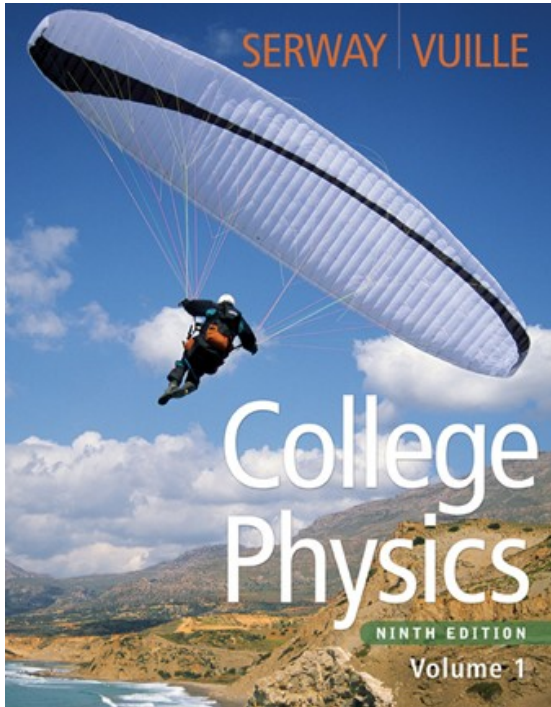




Raymond A. Serway
Chris Vuille

Chapter Nine

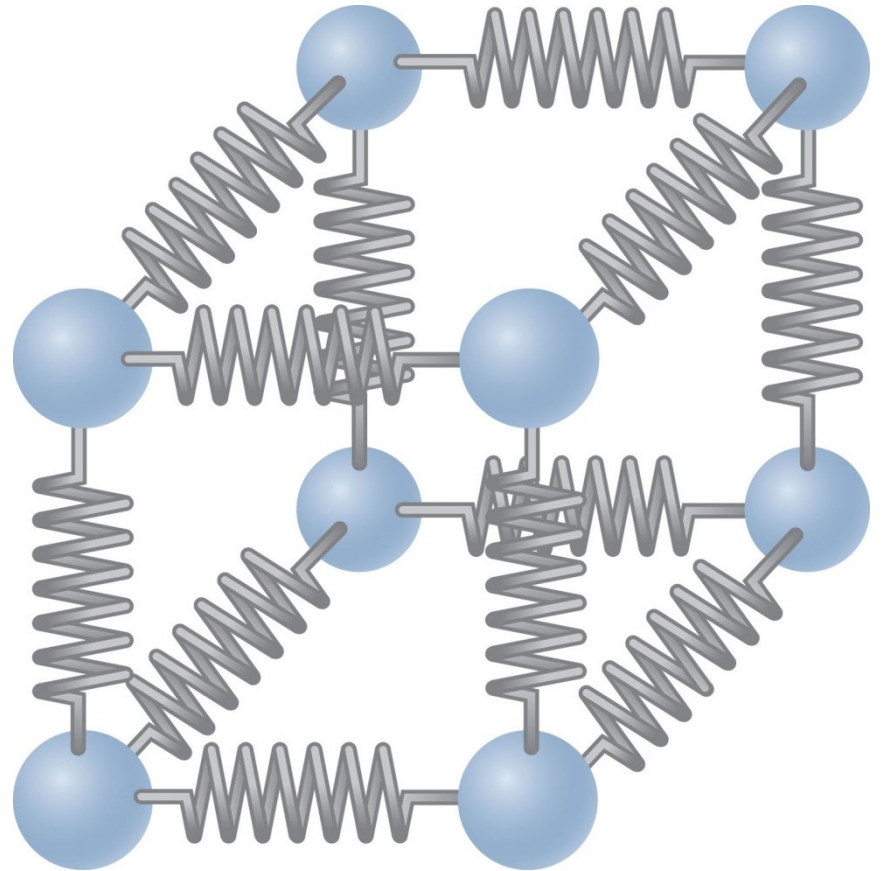
Solids and Fluids

States of Matter

- Solid, liquid, gas
 - Predominate on Earth
- Plasma
 - Predominates in the universe
- This chapter introduces basic properties of solids and liquids
 - Includes some properties of gases

Solids

- Have definite volume
- Have definite shape
- Molecules are held in specific locations
 - By electrical forces
- Vibrate about equilibrium positions
- Can be modeled as springs connecting molecules

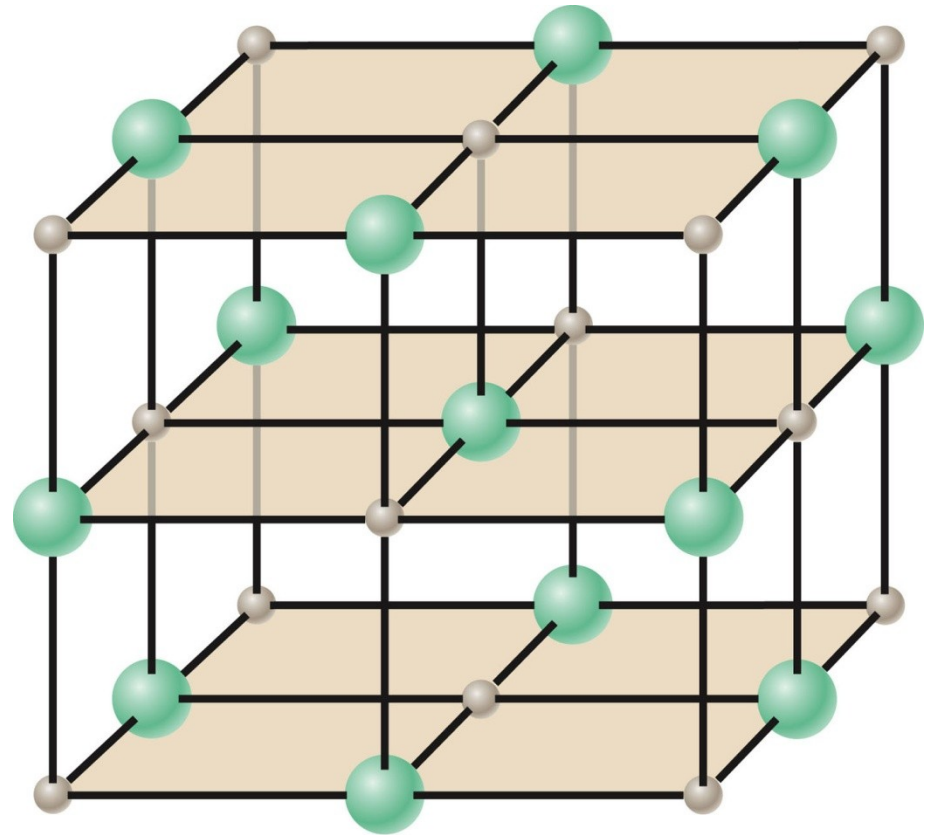


More About Solids

- External forces can be applied to the solid and compress the material
 - In the model, the springs would be compressed
- When the force is removed, the solid returns to its original shape and size
 - This property is called elasticity

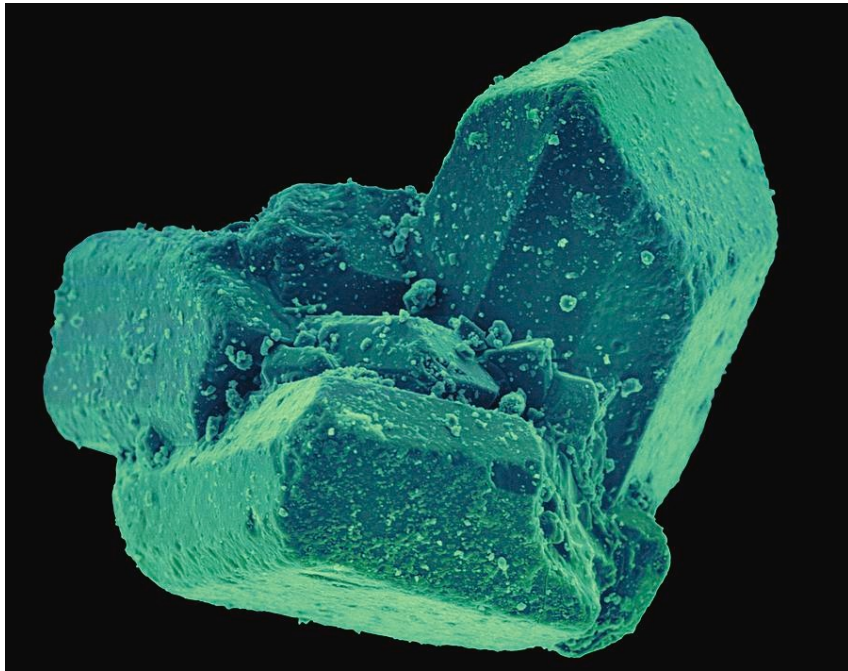
Crystalline Solid

- Atoms have an ordered structure
- This example is salt
 - Gray spheres represent Na^+ ions
 - Green spheres represent Cl^- ions



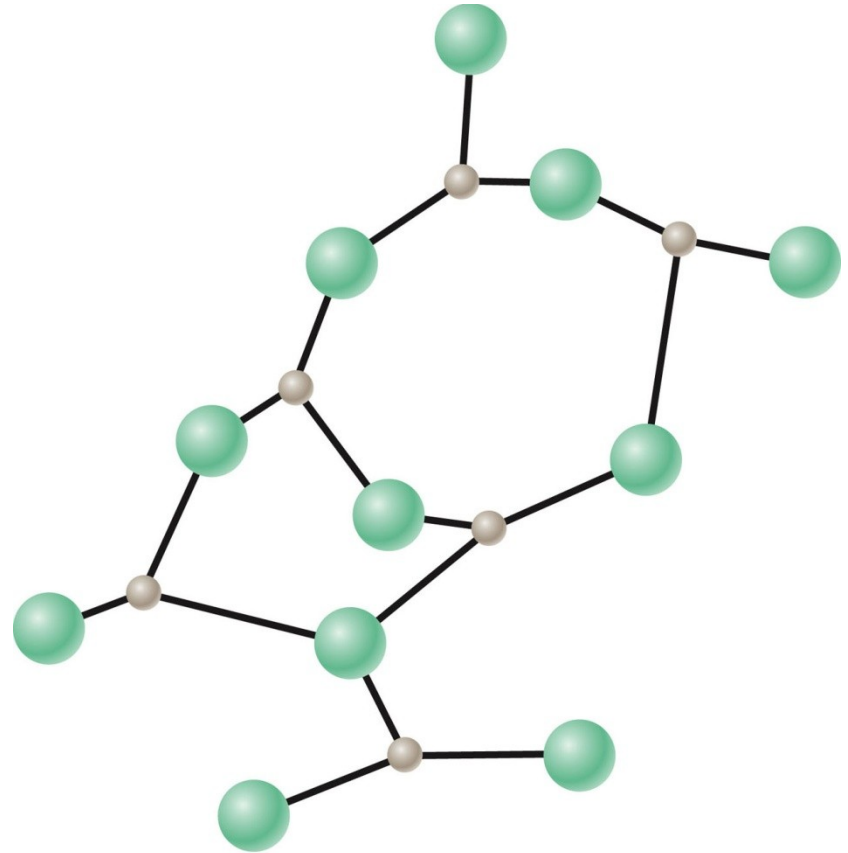
crystals



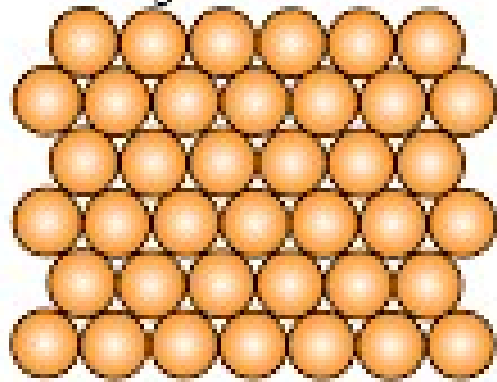


Amorphous Solid

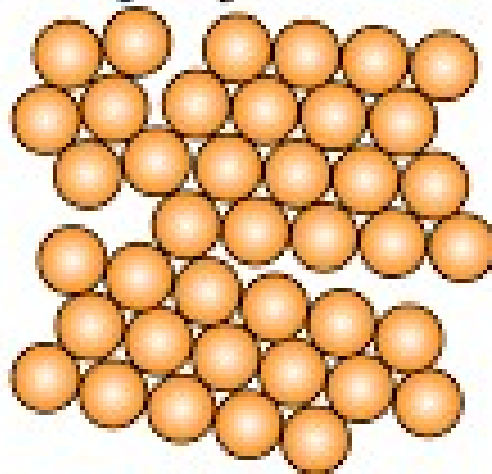
- Atoms are arranged almost randomly
- Examples include glass



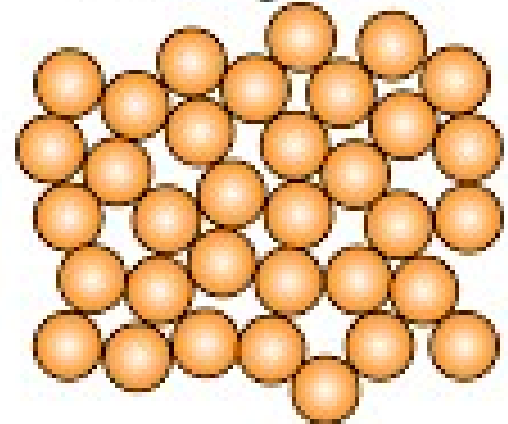
Crystalline



Polycrystalline

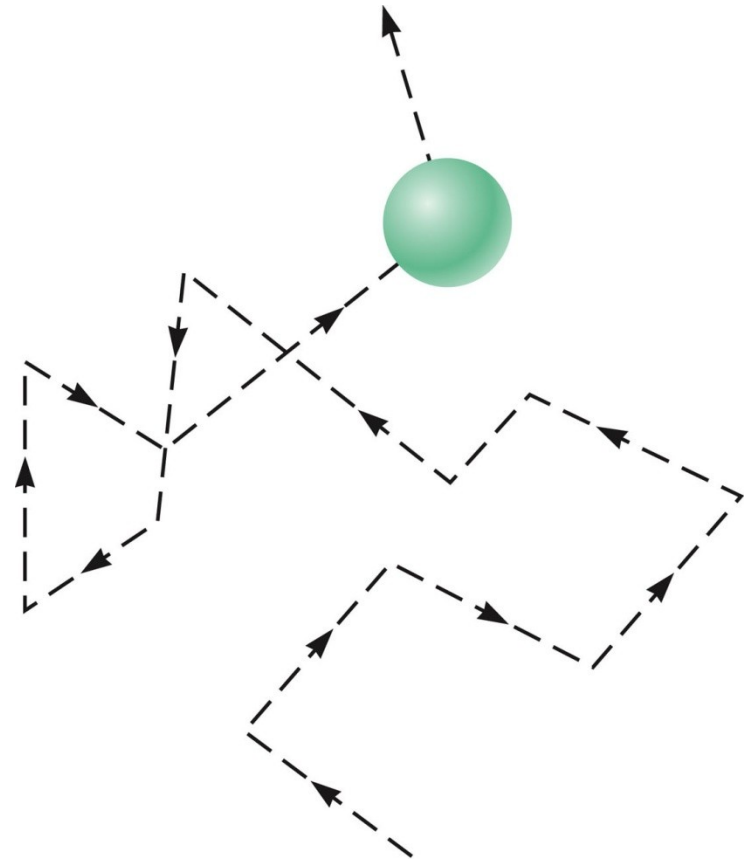


Amorphous



Liquid

- Has a definite volume
- No definite shape
- Exists at a higher temperature than solids
- The molecules “wander” through the liquid in a random fashion
 - The intermolecular forces are not strong enough to keep the molecules in a fixed position



Gas

- Has no definite volume
- Has no definite shape
- Molecules are in constant random motion
- The molecules exert only weak forces on each other
- Average distance between molecules is large compared to the size of the molecules

Plasma

- Gas heated to a very high temperature
- Many of the electrons are freed from the nucleus
- Result is a collection of free, electrically charged ions
- Long-range electrical and magnetic forces allow interactions within the plasma
- Plasmas exist inside stars

Types of Matter

- Normal matter
 - About 5% of total matter
- Dark matter
 - Affects the motion of stars in galaxies
 - May be as much as 25% of total matter
- Dark energy
 - Accounts for acceleration of the expansion of the universe
 - May be as much as 70% of all matter

Density

- The density of a substance of uniform composition is defined as its mass per unit volume:

$$\rho \equiv \frac{m}{V}$$

- SI unit: kg/m³ (SI)
 - Often see g/cm³ (cgs)
- 1 g/cm³ = 1000 kg/m³

Density, cont.

- See table 9.1 for the densities of some common substances
- The densities of most liquids and solids vary slightly with changes in temperature and pressure
- Densities of gases vary greatly with changes in temperature and pressure
- The higher normal densities of solids and liquids compared to gases imply that the average spacing between molecules in a gas is about 10 times greater than the solid or liquid

Density

$$\rho \equiv \frac{M}{V} \quad \text{SI unit: kilogram per meter cubed (kg/m}^3\text{)}$$

Table 9.1 Densities of Some Common Substances

Substance	ρ (kg/m ³) ^a	Substance	ρ (kg/m ³) ^a
Ice	0.917×10^3	Water	1.00×10^3
Aluminum	2.70×10^3	Glycerin	1.26×10^3
Iron	7.86×10^3	Ethyl alcohol	0.806×10^3
Copper	8.92×10^3	Benzene	0.879×10^3
Silver	10.5×10^3	Mercury	13.6×10^3
Lead	11.3×10^3	Air	1.29
Gold	19.3×10^3	Oxygen	1.43
Platinum	21.4×10^3	Hydrogen	8.99×10^{-2}
Uranium	18.7×10^3	Helium	1.79×10^{-1}

^aAll values are at standard atmospheric temperature and pressure (STP), defined as 0°C (273 K) and 1 atm (1.013 × 10⁵ Pa). To convert to grams per cubic centimeter, multiply by 10⁻³.

Specific Gravity

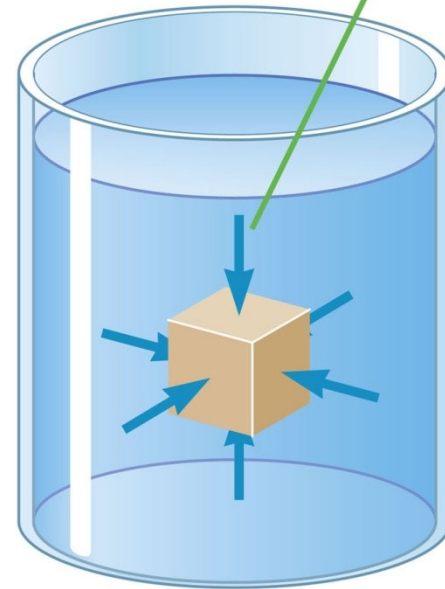
- The *specific gravity* of a substance is the ratio of its density to the density of water at 4° C
 - The density of water at 4° C is 1000 kg/m³
- Specific gravity is a dimensionless quantity

Pressure

- The force exerted by a fluid on a submerged object at any point is perpendicular to the surface of the object
- The average pressure P is the force divided by the area

$$P \equiv \frac{F}{A} \text{ in Pa} = \frac{\text{N}}{\text{m}^2}$$

The force exerted by a fluid on a submerged object at any point is perpendicular to the surface and increases with depth.



Deformation of Solids

- All objects are deformable
- It is possible to change the shape or size (or both) of an object through the application of external forces
- When the forces are removed, the object tends to its original shape
 - An object undergoing this type of deformation exhibits *elastic behavior*

Elastic Properties

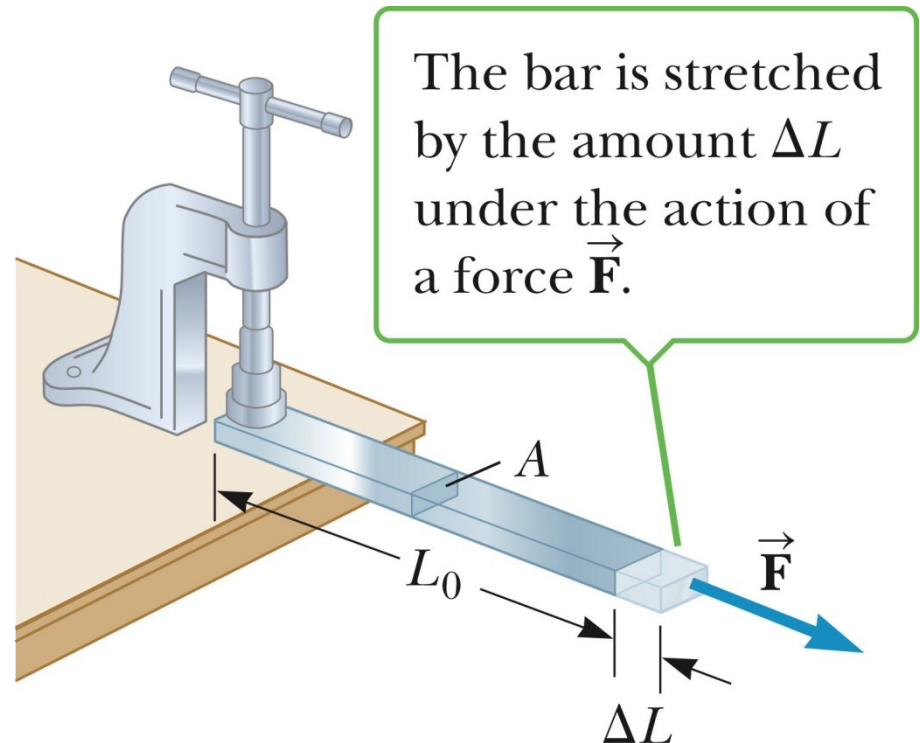
- ***Stress*** is the force per unit area causing the deformation
- ***Strain*** is a measure of the amount of deformation
- The ***elastic modulus*** is the constant of proportionality between stress and strain
 - For sufficiently small stresses, the stress is directly proportional to the strain
 - The constant of proportionality depends on the material being deformed and the nature of the deformation

Elastic Modulus

- stress = elastic modulus x strain
- The elastic modulus can be thought of as the stiffness of the material
 - A material with a large elastic modulus is very stiff and difficult to deform
 - Analogous to the spring constant

Young's Modulus: Elasticity in Length

- The bar is *stressed*
 - Its length is greater than L_0
 - The external force is balanced by internal forces
- Tensile stress is the ratio of the external force to the cross-sectional area
 - Tensile is because the bar is under tension



Young's Modulus, cont.

- SI unit of stress is Pascal, Pa
 - $1 \text{ Pa} = 1 \text{ N/m}^2$
- The tensile strain is the ratio of the change in length to the original length
 - Strain is dimensionless
- The elastic modulus is called *Young's modulus*

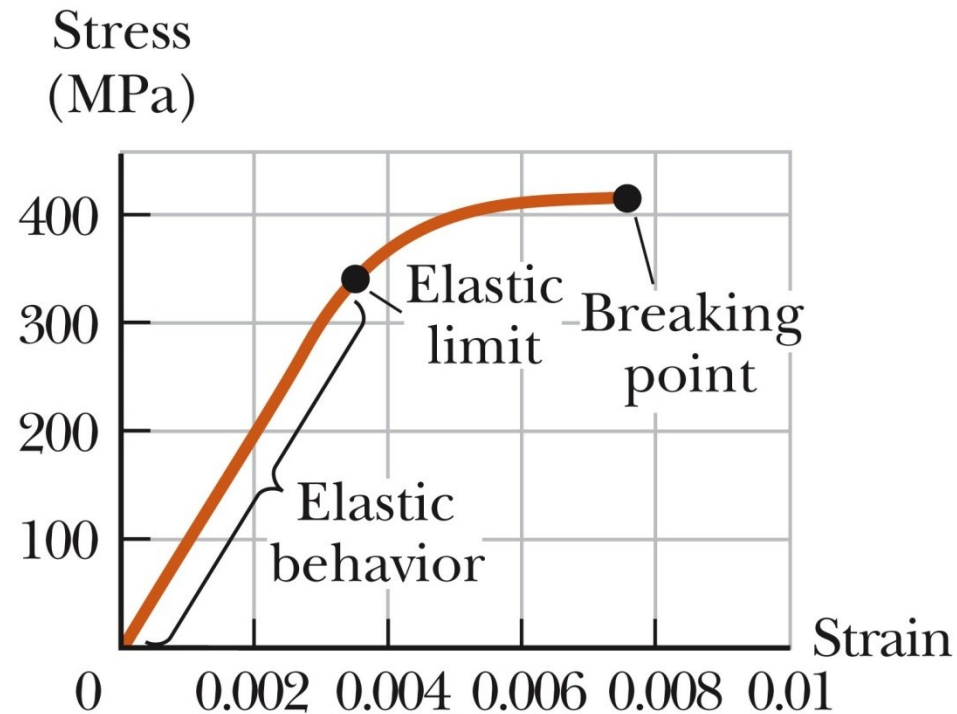
$$\frac{F}{A} = Y \frac{\Delta L}{L_o}$$

Young's Modulus, final

- Young's modulus applies to a stress of either tension or compression
- Experiments show:
 - The change in length for a fixed external force is proportional to the original length
 - The force necessary to produce a given strain is proportional to the cross-sectional area

Elastic Behavior Graph

- It is possible to exceed the *elastic limit* of the material
 - No longer directly proportional
 - Ordinarily does not return to its original length
- If stress continues, it surpasses its *ultimate strength*
 - The ultimate strength is the greatest stress the object can withstand without breaking



Breaking

- The **breaking point**
 - For a brittle material, the breaking point is just beyond its ultimate strength
 - For a ductile material, after passing the ultimate strength the material thins and stretches at a lower stress level before breaking

Moduli Values

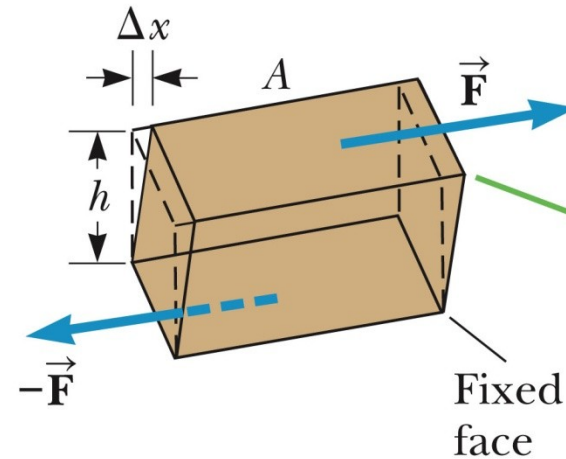
TABLE 12.1

Typical Values for Elastic Moduli

Substance	Young's Modulus (N/m ²)	Shear Modulus (N/m ²)	Bulk Modulus (N/m ²)
Tungsten	35×10^{10}	14×10^{10}	20×10^{10}
Steel	20×10^{10}	8.4×10^{10}	6×10^{10}
Copper	11×10^{10}	4.2×10^{10}	14×10^{10}
Brass	9.1×10^{10}	3.5×10^{10}	6.1×10^{10}
Aluminum	7.0×10^{10}	2.5×10^{10}	7.0×10^{10}
Glass	$6.5\text{--}7.8 \times 10^{10}$	$2.6\text{--}3.2 \times 10^{10}$	$5.0\text{--}5.5 \times 10^{10}$
Quartz	5.6×10^{10}	2.6×10^{10}	2.7×10^{10}
Water	—	—	0.21×10^{10}
Mercury	—	—	2.8×10^{10}

Shear Modulus: Elasticity of Shape

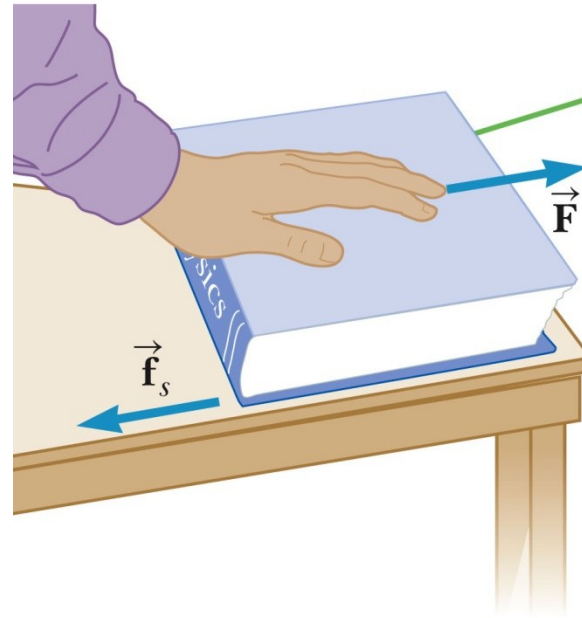
- Forces may be parallel to one of the object's faces
- The stress is called a shear stress
 - Defined as the ratio of the magnitude of the parallel force to the area of the face being sheared
- The shear strain is the ratio of the horizontal displacement and the height of the object
- The shear modulus is S



The shear stress displaces the top face of the block to the right relative the bottom.

Shear Modulus, Equations

- $$\text{shear stress} = \frac{F}{A}$$
$$\text{shear strain} = \frac{\Delta x}{h}$$
$$\frac{F}{A} = S \frac{\Delta x}{h}$$
- S is the shear modulus
- A material having a large shear modulus is difficult to bend



The shear stress displaces the front cover of the book to the right relative the back cover.

Shear Modulus, final

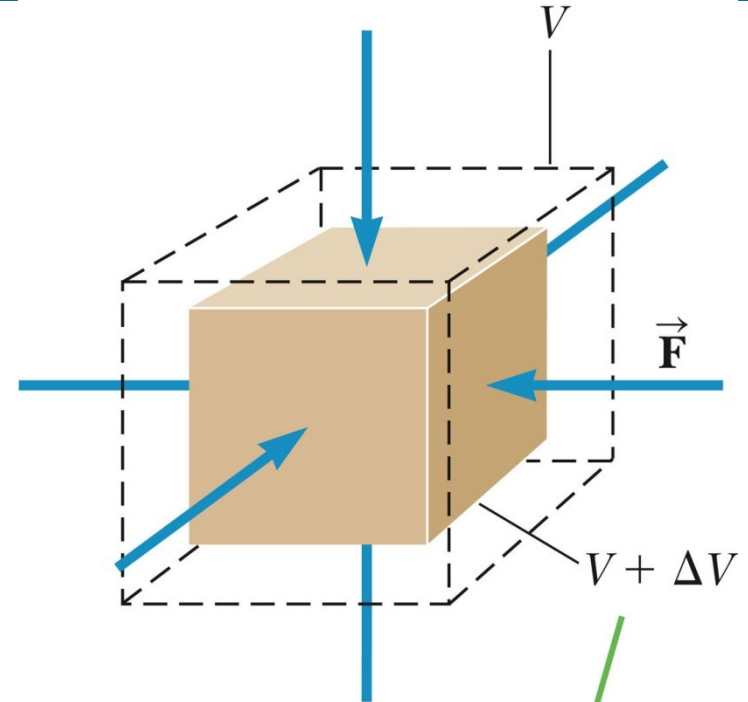
- There is no volume change in this type of deformation
- Remember the force is parallel to the cross-sectional area
 - In tensile stress, the force is perpendicular to the cross-sectional area

Bulk Modulus: Volume Elasticity

- Bulk modulus characterizes the response of an object to uniform squeezing
 - Suppose the forces are perpendicular to, and act on, all the surfaces
 - Example: when an object is immersed in a fluid
- The object undergoes a change in volume without a change in shape

Bulk Modulus, cont.

- Volume stress, ΔP , is the ratio of the change in the magnitude of the applied force to the surface area
 - This is also a change in pressure
- The volume strain is equal to the ratio of the change in volume to the original volume



Under uniform bulk stress, the cube shrinks in size without changing shape.

Bulk Modulus, final

$$\Delta P = -B \frac{\Delta V}{V}$$

- A material with a large bulk modulus is difficult to compress
- The negative sign is included since an increase in pressure will produce a decrease in volume
 - B is always positive
- The *compressibility* is the reciprocal of the bulk modulus

Notes on Moduli

- Solids have Young's, Bulk, and Shear moduli
- Liquids have only bulk moduli, they will not undergo a shearing or tensile stress
 - The liquid would flow instead

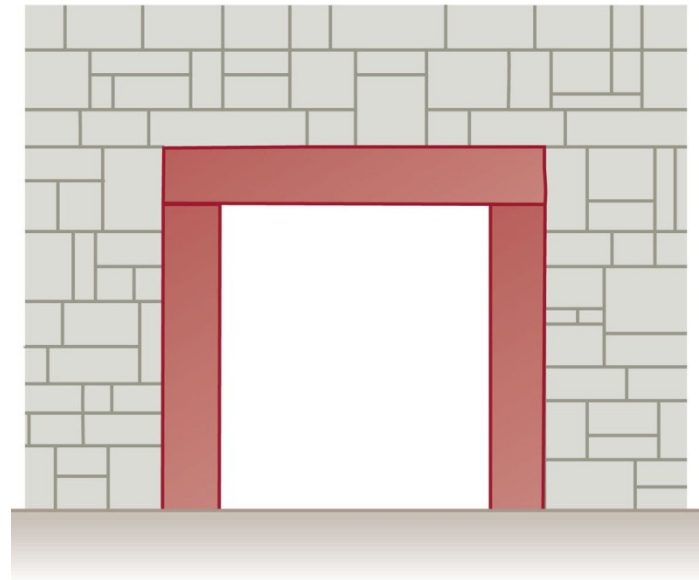
Ultimate Strength of Materials

- The *ultimate strength* of a material is the maximum force per unit area the material can withstand before it breaks or fractures
- Some materials are stronger in compression than in tension

Post and Beam Arches

- A horizontal beam is supported by two columns
- Used in Greek temples
- Columns are closely spaced
 - Limited length of available stones
 - Low ultimate tensile strength of sagging stone beams

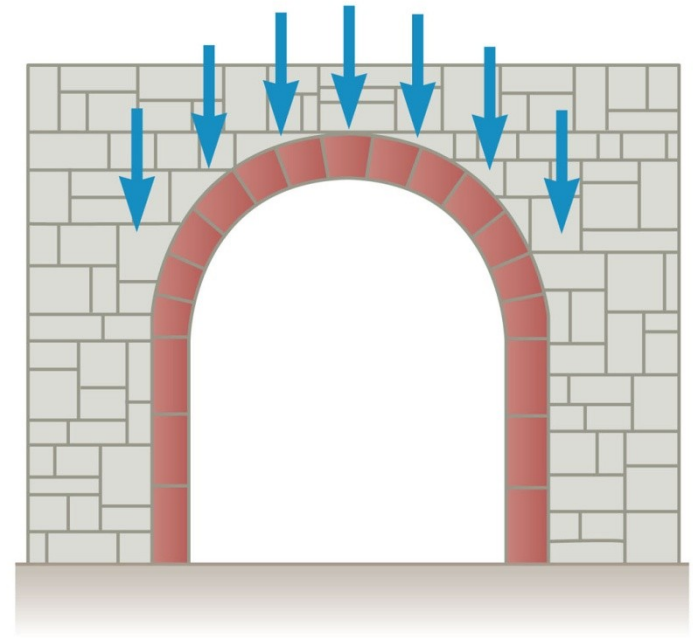
Post and beam



Semicircular Arch

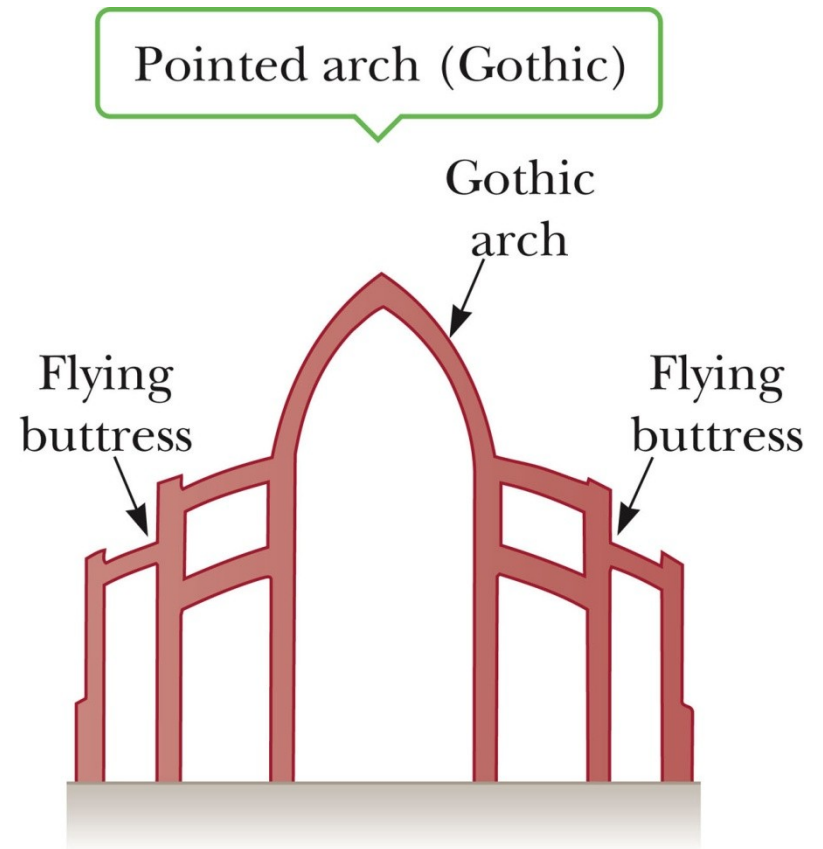
- Developed by the Romans
- Allows a wide roof span on narrow supporting columns
- Stability depends upon the compression of the wedge-shaped stones

Semicircular arch (Roman)



Gothic Arch

- First used in Europe in the 12th century
- Extremely high
- The flying buttresses are needed to prevent the spreading of the arch supported by the tall, narrow columns

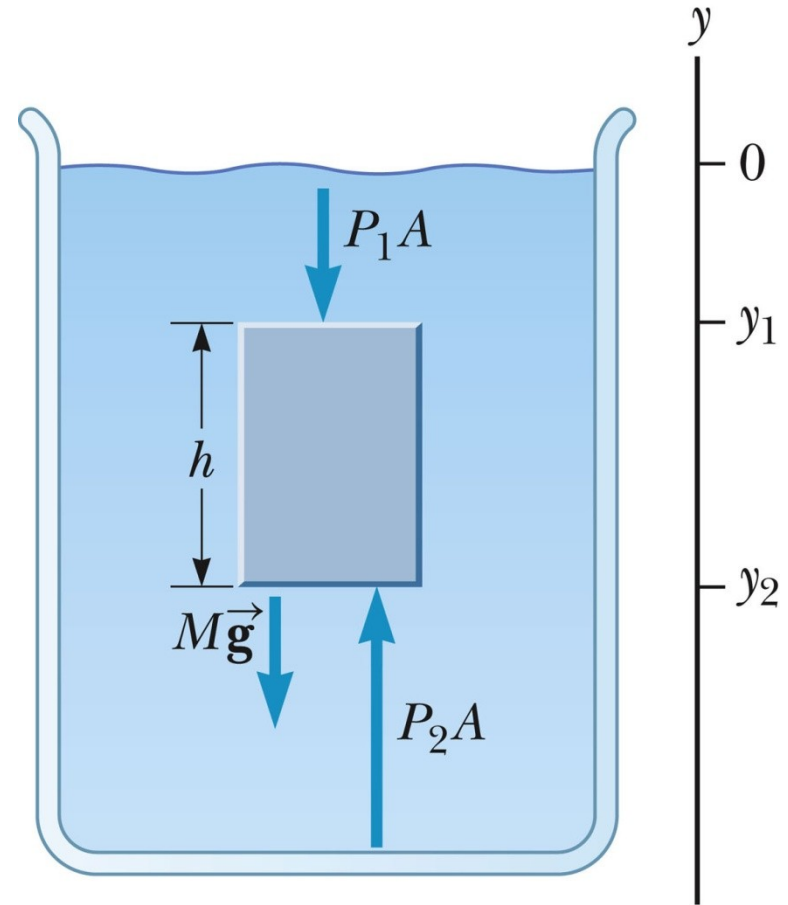


Variation of Pressure with Depth

- If a fluid is at rest in a container, all portions of the fluid must be in static equilibrium
- All points at the same depth must be at the same pressure
 - Otherwise, the fluid would not be in equilibrium
 - The fluid would flow from the higher pressure region to the lower pressure region

Pressure and Depth

- Examine the darker region, assumed to be a fluid
 - It has a cross-sectional area A
 - Extends to a depth h below the surface
- Three external forces act on the region



Pressure and Depth equation

- $P = P_o + \rho gh$
- P_o is normal atmospheric pressure
 - $1.013 \times 10^5 \text{ Pa} = 14.7 \text{ lb/in.}^2$
- The pressure does not depend upon the shape of the container



Pascal's Principle

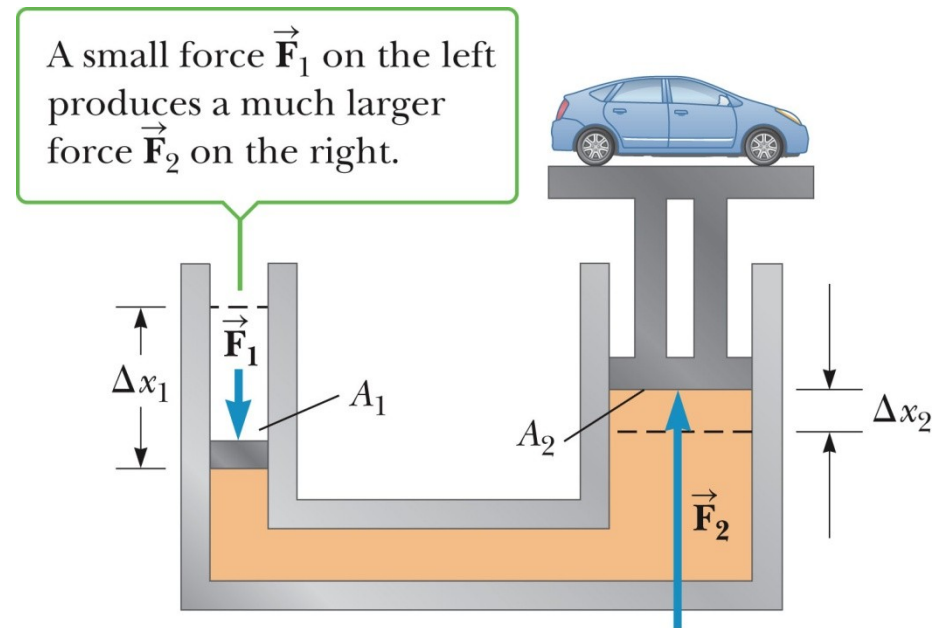
- A change in pressure applied to an enclosed fluid is transmitted undiminished to every point of the fluid and to the walls of the container.
 - First recognized by Blaise Pascal, a French scientist (1623 – 1662)

Pascal's Principle, cont

- The hydraulic press is an important application of Pascal's Principle

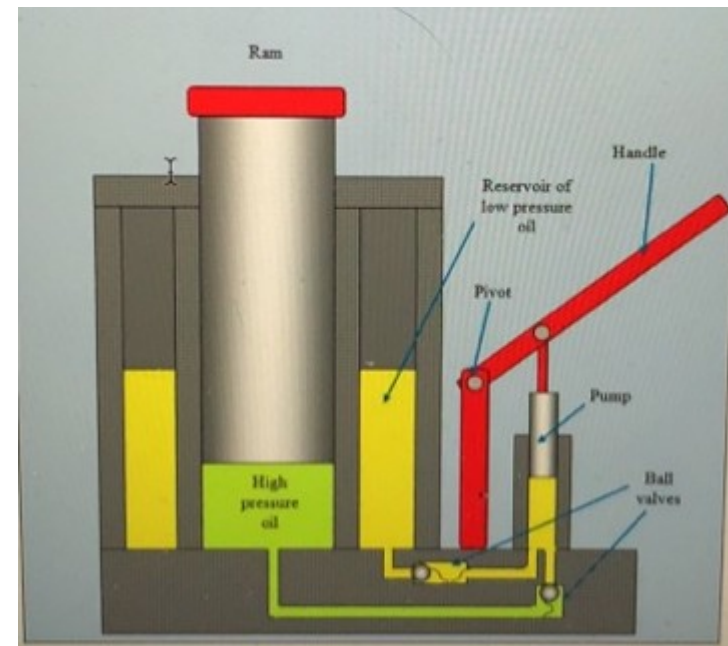
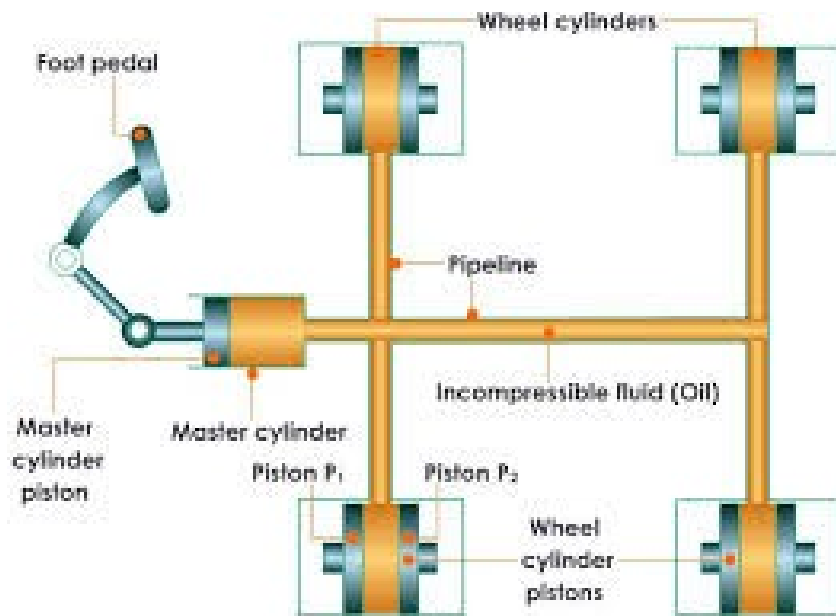
$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

- Also used in hydraulic brakes, forklifts, car lifts, etc.

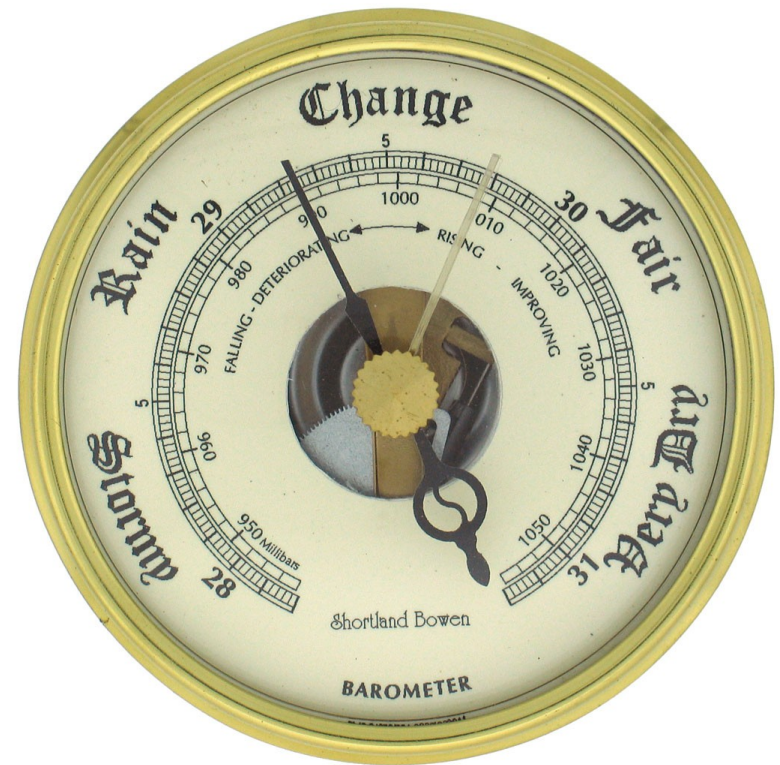


Hydraulic Brake and Jack

- Jack

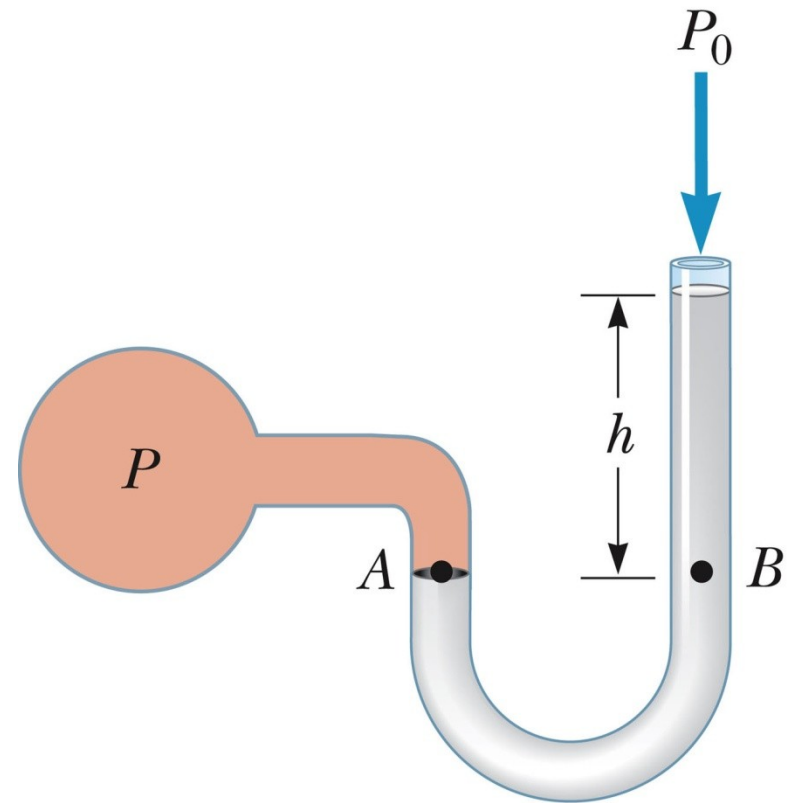


Pressure Measurement



Pressure Measurements: Manometer

- One end of the U-shaped tube is open to the atmosphere
- The other end is connected to the pressure to be measured
- If P in the system is greater than atmospheric pressure, h is positive
 - If less, then h is negative

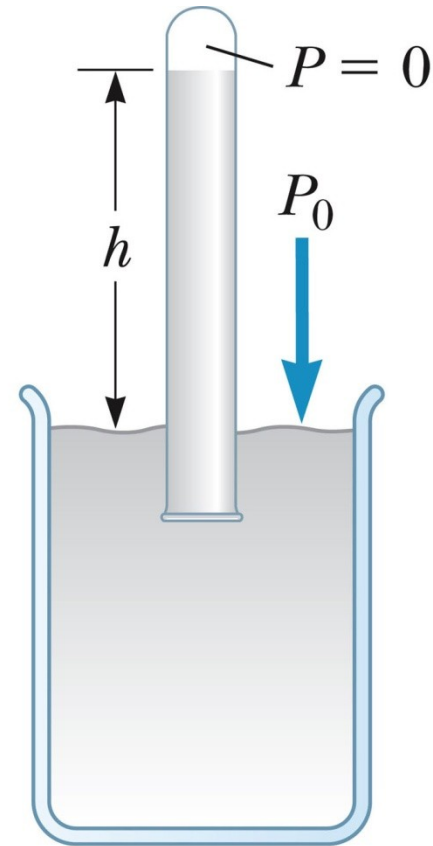


Absolute vs. Gauge Pressure

- The pressure P is called the ***absolute*** pressure
 - Remember, $P = P_o + \rho gh$
- $P - P_o = \rho gh$ is the ***gauge*** pressure

Pressure Measurements: Barometer

- Invented by Torricelli (1608 – 1647)
- A long closed tube is filled with mercury and inverted in a dish of mercury
- Measures atmospheric pressure as ρgh

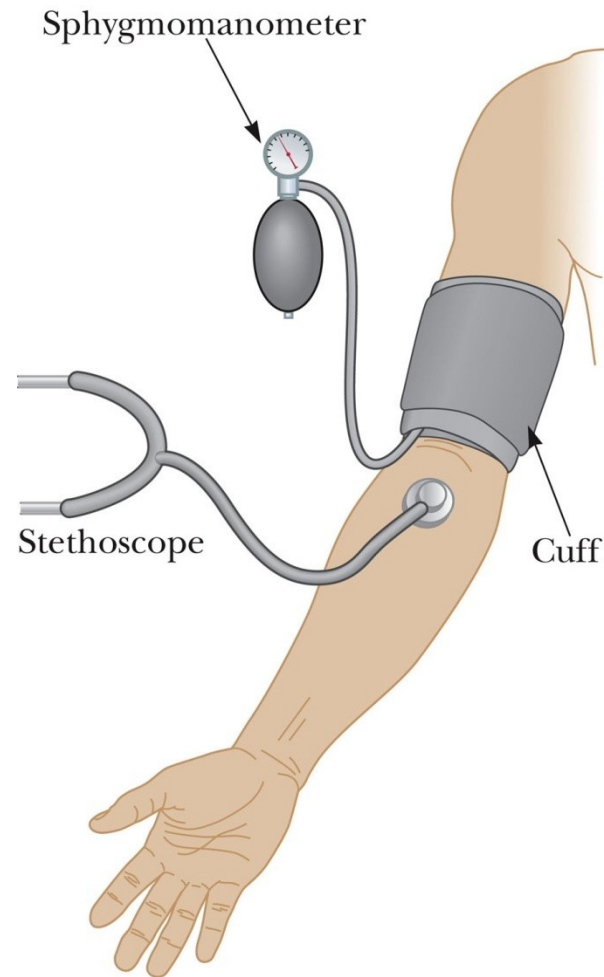


Pressure Values in Various Units

- One atmosphere of pressure is defined as the pressure equivalent to a column of mercury exactly 0.76 m tall at 0° C where $g = 9.806\,65\text{ m/s}^2$
- One atmosphere (1 atm) =
 - 76.0 cm of mercury
 - $1.013 \times 10^5\text{ Pa}$
 - 14.7 lb/in^2

Blood Pressure

- Blood pressure is measured with a special type of manometer called a *sphygmomanometer*
- Pressure is measured in mm of mercury



Archimedes

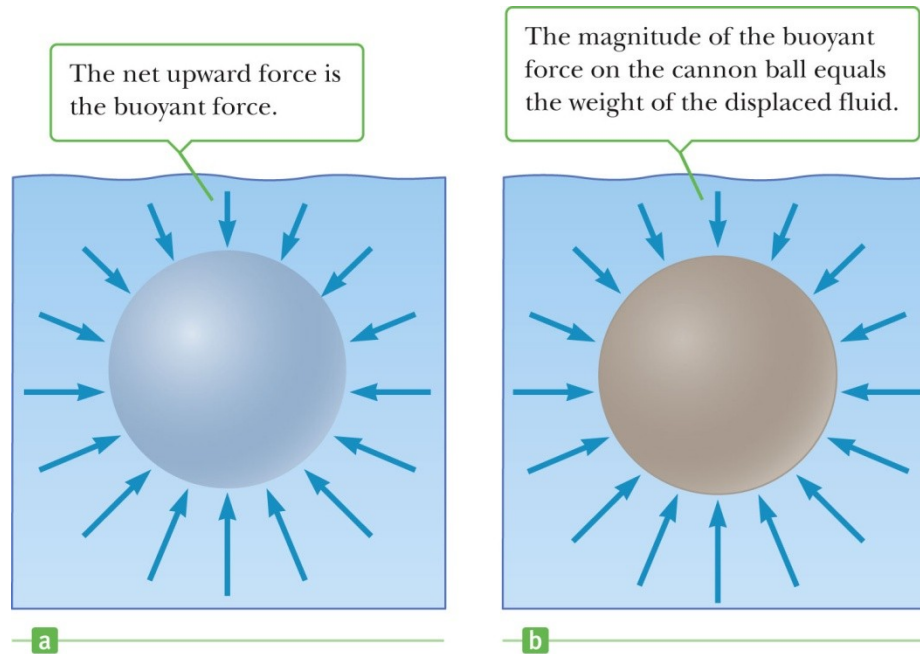
- 287 – 212 BC
- Greek mathematician, physicist, and engineer
- Buoyant force
- Inventor



Archimedes' Principle

- Any object completely or partially submerged in a fluid is buoyed up by a force whose magnitude is equal to the weight of the fluid displaced by the object.
- *See Hewitt Drew it No 60: Archimedes*

Buoyant Force



- The upward force is called the *buoyant force*
- The physical cause of the buoyant force is the pressure difference between the top and the bottom of the object

Buoyant Force, cont.

- The magnitude of the buoyant force always equals the weight of the displaced fluid

$$B = \rho_{fluid} V_{fluid} g = w_{fluid}$$

- The buoyant force is the same for a totally submerged object of any shape or density

Buoyant Force, final

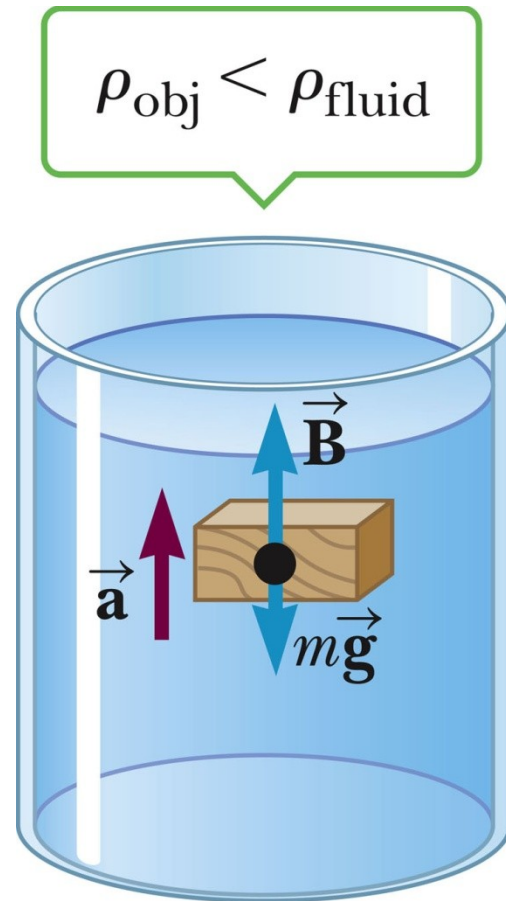
- The buoyant force is exerted by the fluid
- Whether an object sinks or floats depends on the relationship between the buoyant force and the weight

Archimedes' Principle: Totally Submerged Object

- The upward buoyant force is $B = \rho_{\text{fluid}} V_{\text{obj}} g$
- The downward gravitational force is $w = mg = \rho_{\text{obj}} V_{\text{obj}} g$
- The net force is $B - w = (\rho_{\text{fluid}} - \rho_{\text{obj}}) V_{\text{obj}} g$

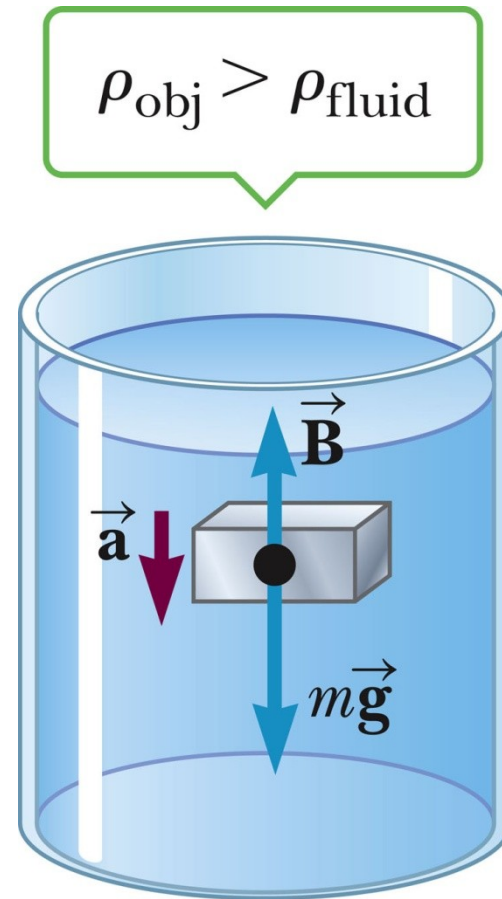
Totally Submerged Object

- The object is less dense than the fluid
- The object experiences a net upward force



Totally Submerged Object, 2

- The object is more dense than the fluid
- The net force is downward
- The object accelerates downward



Archimedes' Principle: Floating Object

- The object is in static equilibrium
- The upward buoyant force is balanced by the downward force of gravity
- Volume of the fluid displaced corresponds to the volume of the object beneath the fluid level

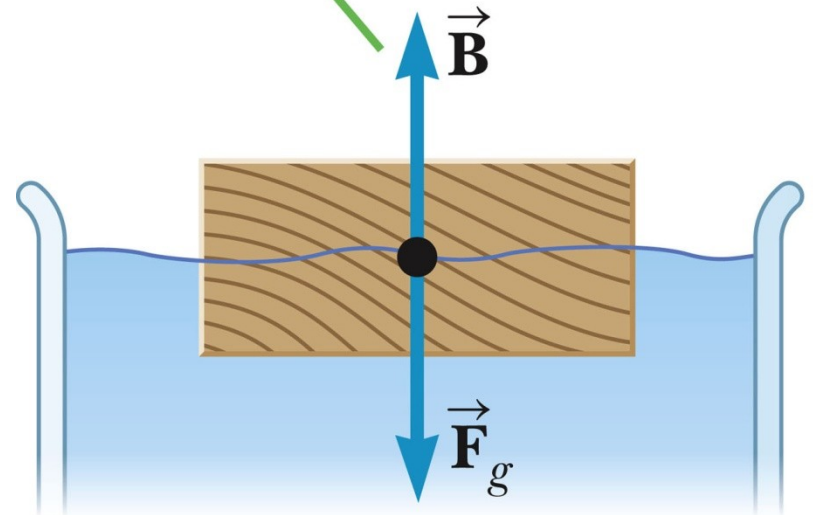
Archimedes' Principle: Floating Object, cont

- The forces balance

$$\begin{aligned}\frac{\rho_{\text{obj}}}{\rho_{\text{fluid}}} &= \frac{V_{\text{fluid displaced}}}{V_{\text{object (total)}}} \\ &= \frac{V_{\text{solid submerged}}}{V_{\text{solid (total)}}}\end{aligned}$$

Neglects the buoyant force
of the air

The two forces are equal
in magnitude and
opposite in direction.

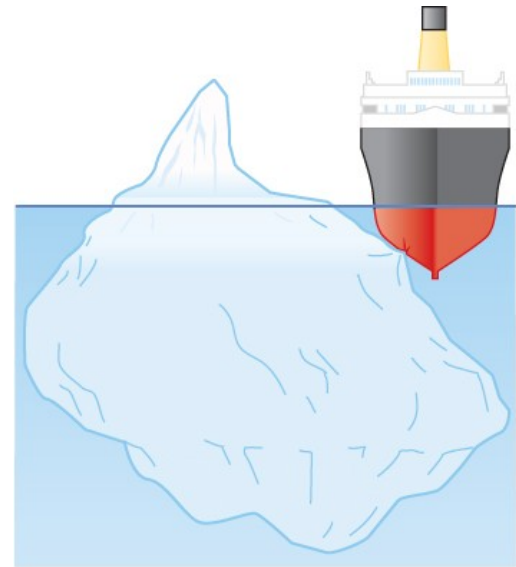


A Titanic Surprise

An iceberg floating in seawater is extremely dangerous because most of the ice is below the surface. This hidden ice can damage a ship that is still a considerable distance from the visible ice. What fraction of the iceberg lies below the water level?



a



b

Fluids in Motion: Streamline Flow

- Streamline flow
 - Every particle that passes a particular point moves exactly along the smooth path followed by particles that passed the point earlier
 - Also called *laminar* flow
- Streamline is the path
 - Different streamlines cannot cross each other
 - The streamline at any point coincides with the direction of fluid velocity at that point

Streamline Flow, Example

- Streamline flow shown around an auto in a wind tunnel



Fluids in Motion: Turbulent Flow

- The flow becomes irregular
 - Exceeds a certain velocity
 - Any condition that causes abrupt changes in velocity
- Eddy currents are a characteristic of turbulent flow

Turbulent Flow, Example

- The smoke first moves in laminar flow at the bottom
- Turbulent flow occurs at the top



Fluid Flow: Viscosity

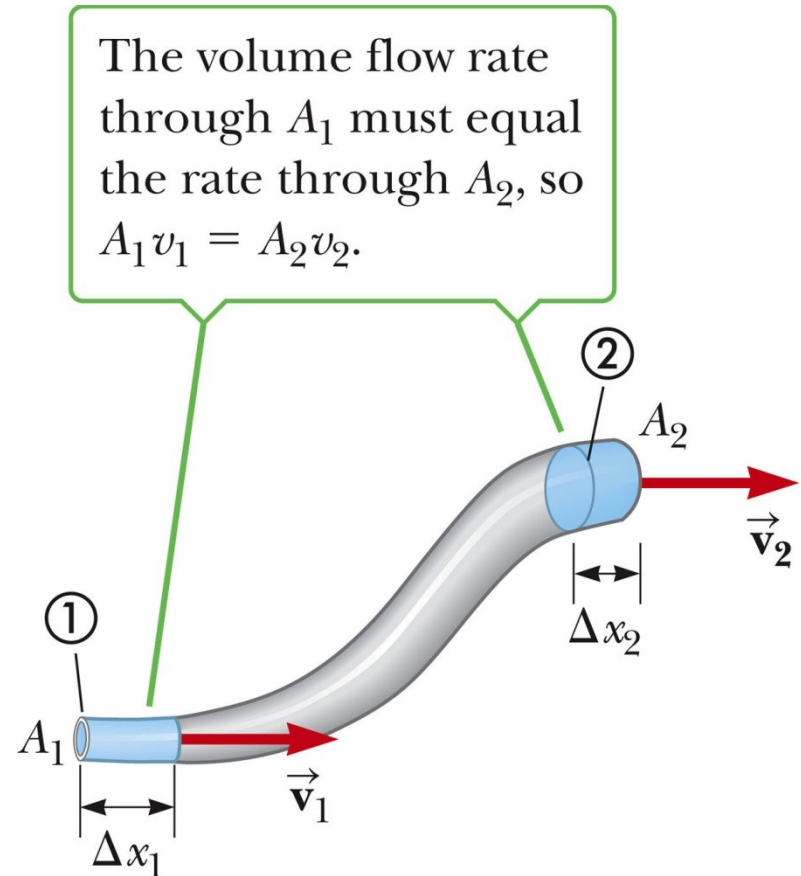
- Viscosity is the degree of internal friction in the fluid
- The internal friction is associated with the resistance between two adjacent layers of the fluid moving relative to each other

Characteristics of an Ideal Fluid

- The fluid is nonviscous
 - There is no internal friction between adjacent layers
- The fluid is incompressible
 - Its density is constant
- The fluid motion is steady
 - The velocity, density, and pressure at each point in the fluid do not change with time
- The fluid moves without turbulence
 - No eddy currents are present
 - The elements have zero angular velocity about its center

Equation of Continuity

- $A_1 v_1 = A_2 v_2$
- The product of the cross-sectional area of a pipe and the fluid speed is a constant
 - Speed is high where the pipe is narrow and speed is low where the pipe has a large diameter
- The product Av is called the flow rate



Equation of Continuity, cont

- The equation is a consequence of conservation of mass and a steady flow
- $A v = \text{constant}$
 - This is equivalent to the fact that the volume of fluid that enters one end of the tube in a given time interval equals the volume of fluid leaving the tube in the same interval
 - Assumes the fluid is incompressible and there are no leaks

Daniel Bernoulli

- 1700 – 1782
- Swiss physicist and mathematician
- Wrote *Hydrodynamica*
- Also did work that was the beginning of the kinetic theory of gases



Bernoulli's Equation

- Relates pressure to fluid speed and elevation
- Bernoulli's equation is a consequence of Conservation of Energy applied to an ideal fluid
- Assumes the fluid is incompressible and nonviscous, and flows in a nonturbulent, steady-state manner

Bernoulli's Equation, cont.

- States that the sum of the pressure, kinetic energy per unit volume, and the potential energy per unit volume has the same value at all points along a streamline

$$P + \frac{1}{2}\rho v^2 + \rho gy = \text{constant}$$

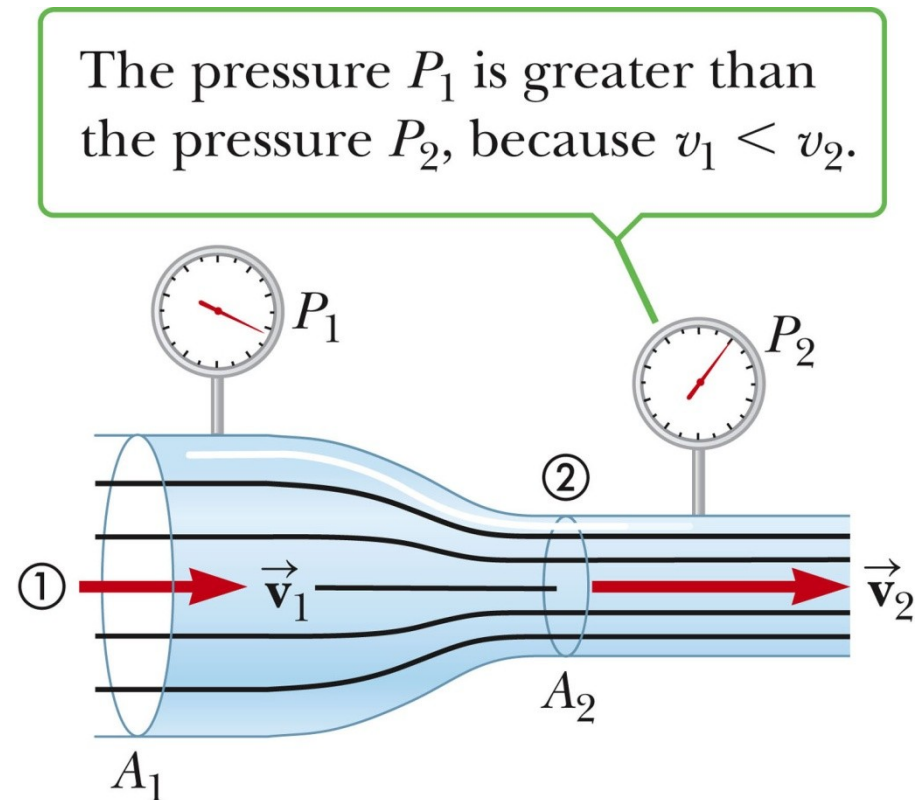
Applications of Fluid Dynamics

Some interesting examples of the results of Bernoulli Equation can be seen here:

<http://www.youtube.com/watch?v=P-xNXrELCmU>

Applications of Bernoulli's Principle: Measuring Speed

- Shows fluid flowing through a horizontal constricted pipe
- Speed changes as diameter changes
- Can be used to measure the speed of the fluid flow
- Swiftly moving fluids exert less pressure than do slowly moving fluids



Applications of Bernoulli's Principle: Venturi Tube

- The height is higher in the constricted area of the tube
- This indicates that the pressure is lower

Pressure is lower at the narrow part of the tube, so the fluid level is higher.

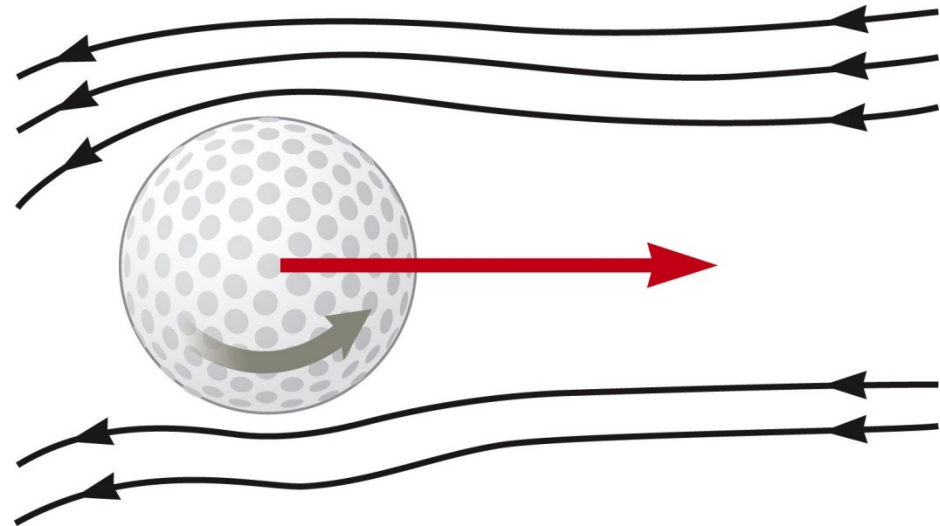


An Object Moving Through a Fluid

- Many common phenomena can be explained by Bernoulli's equation
 - At least partially
- In general, an object moving through a fluid is acted upon by a net upward force as the result of any effect that causes the fluid to change its direction as it flows past the object
- Swiftly moving fluids exert less pressure than do slowing moving fluids

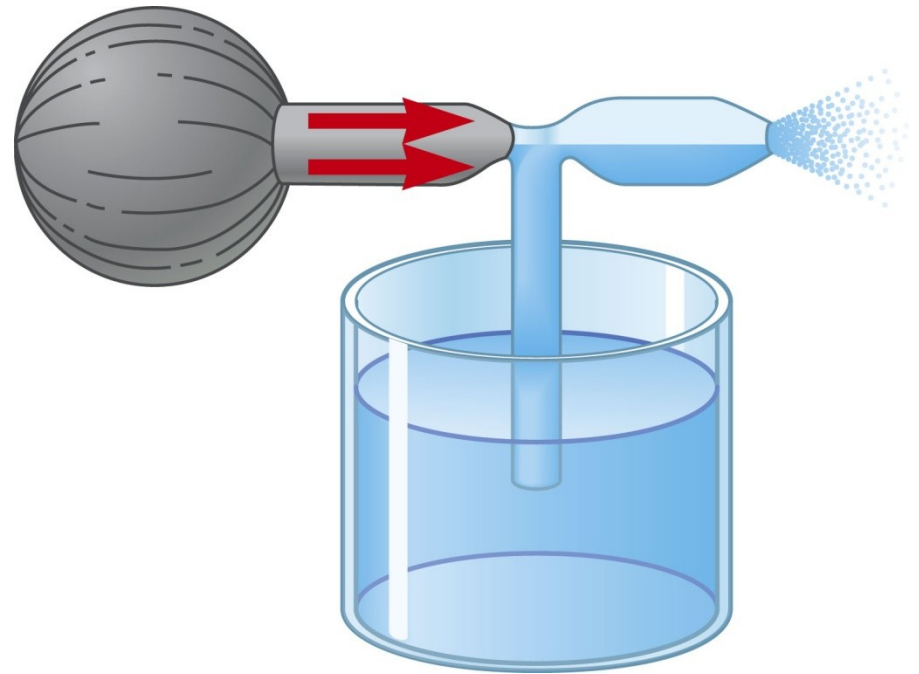
Application – Golf Ball

- The dimples in the golf ball help move air along its surface
- The ball pushes the air down
- Newton's Third Law tells us the air must push up on the ball
- The spinning ball travels farther than if it were not spinning



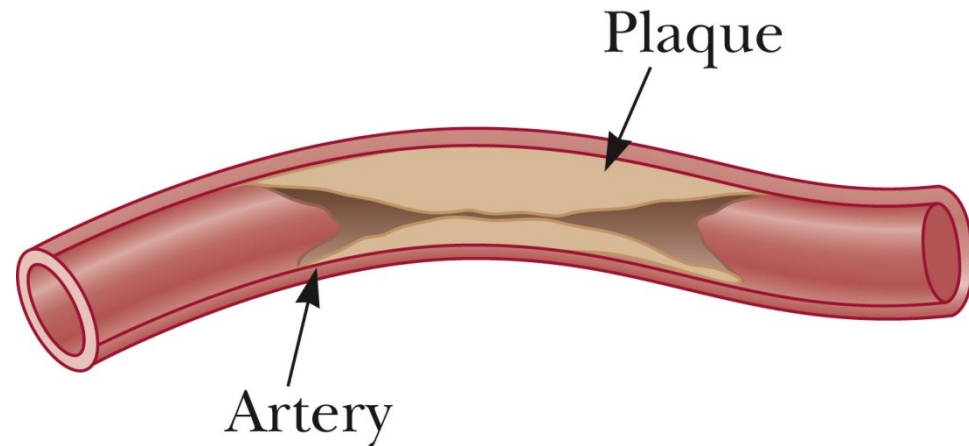
Application – Atomizer

- A stream of air passing over an open tube reduces the pressure above the tube
- The liquid rises into the airstream
- The liquid is then dispersed into a fine spray of droplets



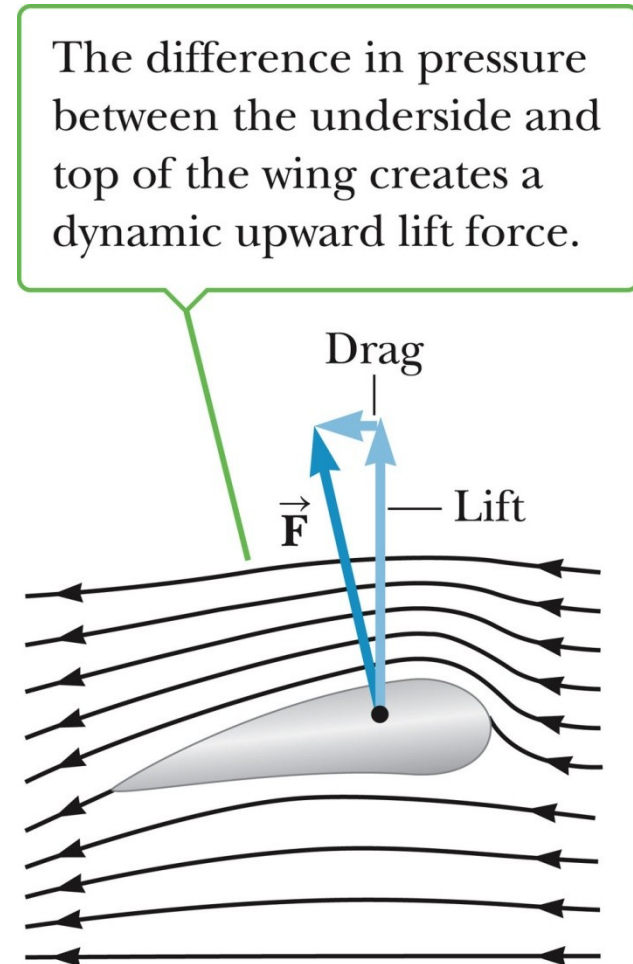
Application – Vascular Flutter

- The artery is constricted as a result of accumulated plaque on its inner walls
- To maintain a constant flow rate, the blood must travel faster than normal
- If the speed is high enough, the blood pressure is low and the artery may collapse



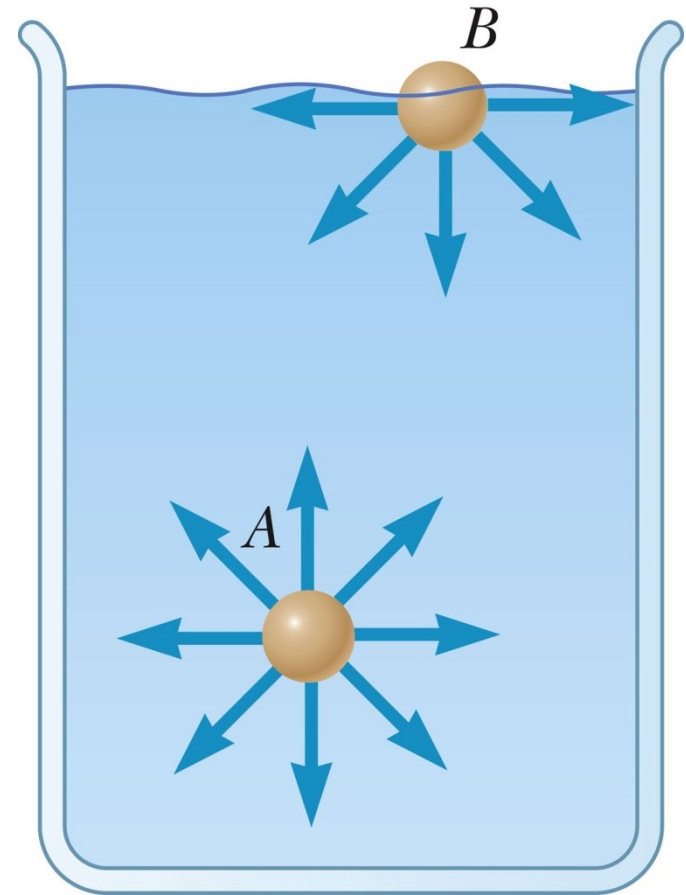
Application – Airplane Wing

- The air speed above the wing is greater than the speed below
- The air pressure above the wing is less than the air pressure below
- There is a net upward force
 - Called *lift*
- Other factors are also involved



Surface Tension

- Net force on molecule A is zero
 - Pulled equally in all directions
- Net force on B is not zero
 - No molecules above to act on it
 - Pulled toward the interior of the fluid

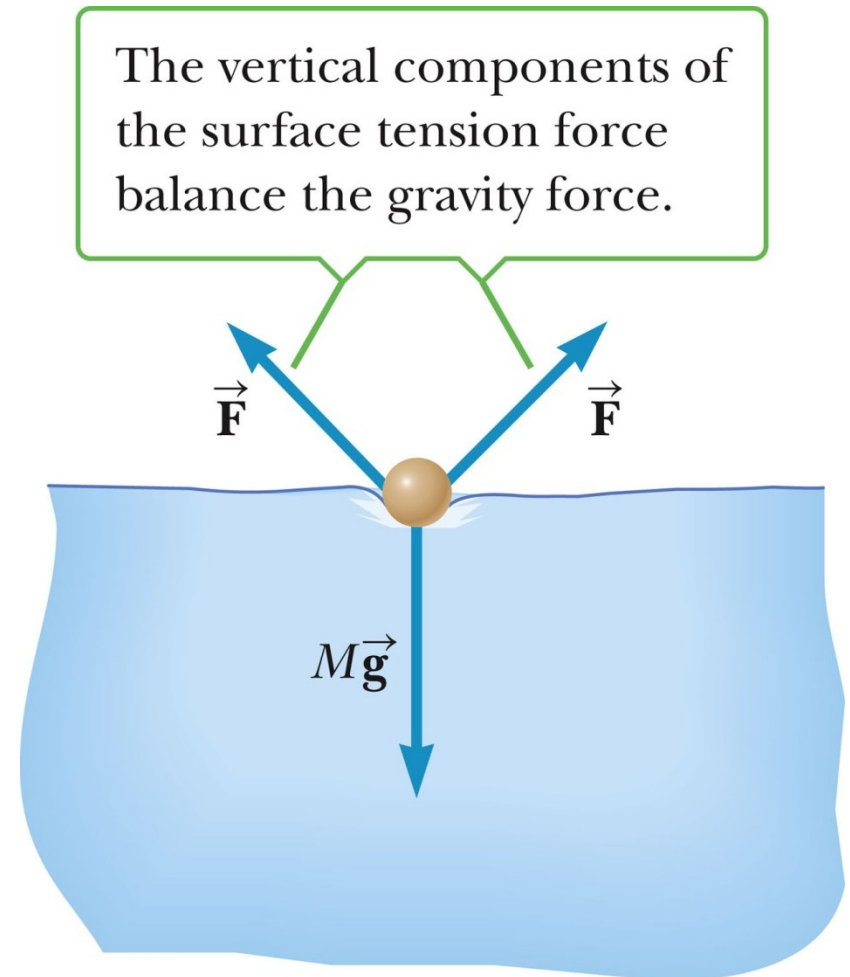


Surface Tension, cont

- The net effect of this pull on all the surface molecules is to make the surface of the liquid contract
- Makes the surface area of the liquid as small as possible
 - Example: Water droplets take on a spherical shape since a sphere has the smallest surface area for a given volume

Surface Tension on a Needle

- Surface tension allows the needle to float, even though the density of the steel in the needle is much higher than the density of the water
- The needle actually rests in a small depression in the liquid surface
- The vertical components of the force balance the weight



Pin, needle and mosquito on water



Surface Tension, Equation

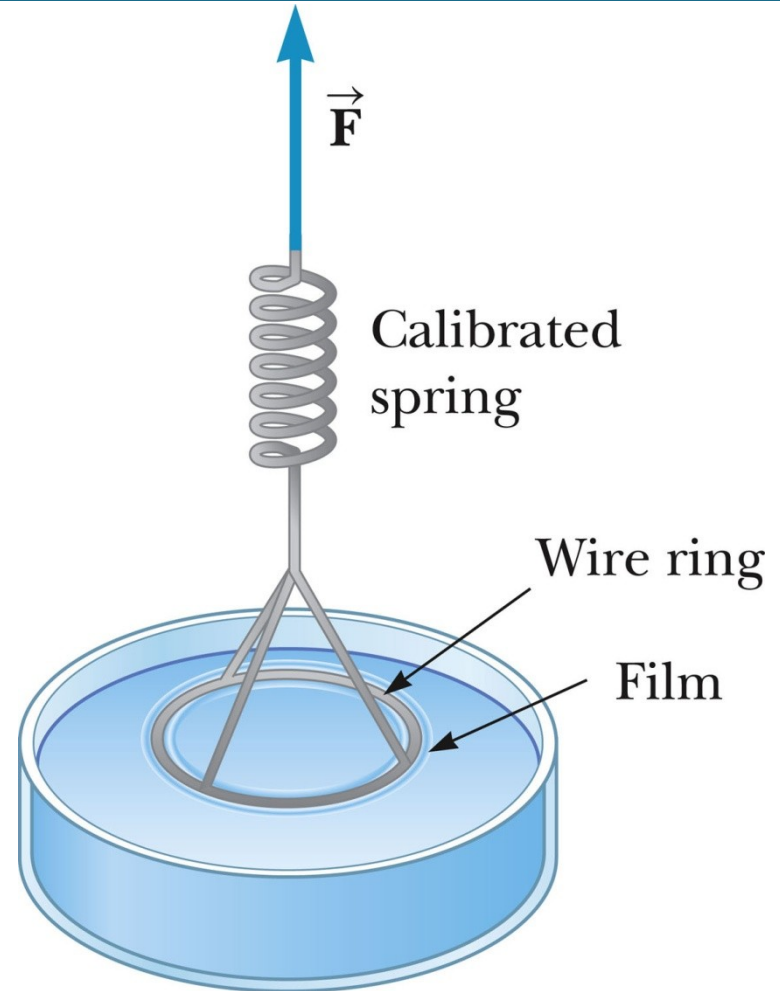
- The surface tension in a film of liquid is defined as the ratio of the magnitude of the surface tension force to the length along which the force acts:

$$\gamma = \frac{F}{L}$$

- SI unit: N/m
- In terms of energy, any equilibrium configuration of an object is one in which the energy is a minimum

Measuring Surface Tension

- The force is measured just as the ring breaks free from the film
- $\gamma = \frac{F}{L}$
- The $2L$ is due to the force being exerted on the inside and outside of the ring



Final Notes About Surface Tension

- The surface tension of liquids decreases with increasing temperature
- Surface tension can be decreased by adding ingredients called *surfactants* to a liquid
 - Detergent is an example

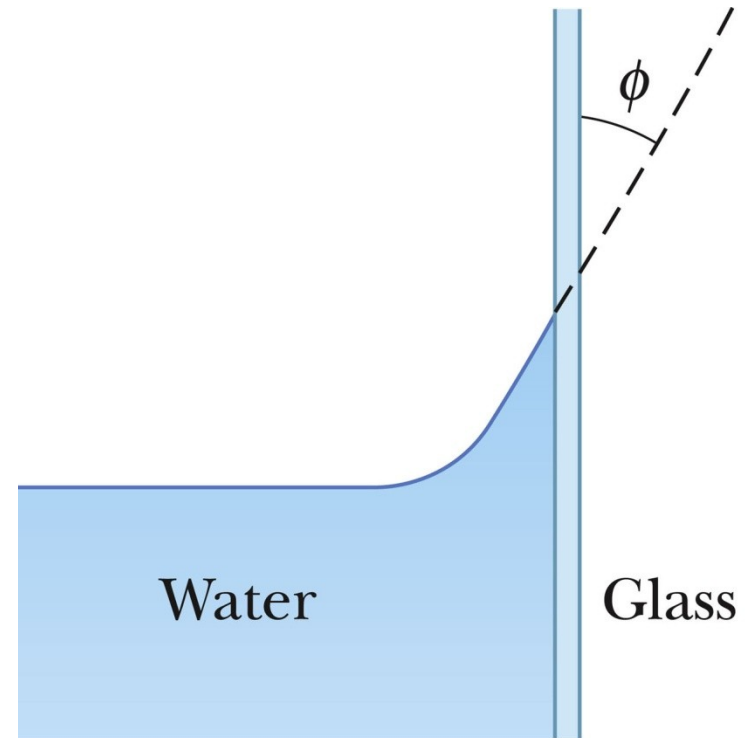
A Closer Look at the Surface of Liquids

- *Cohesive forces* are forces between like molecules
- *Adhesive forces* are forces between unlike molecules
- The shape of the surface depends upon the relative size of the cohesive and adhesive forces

Liquids in Contact with a Solid Surface

– Case 1

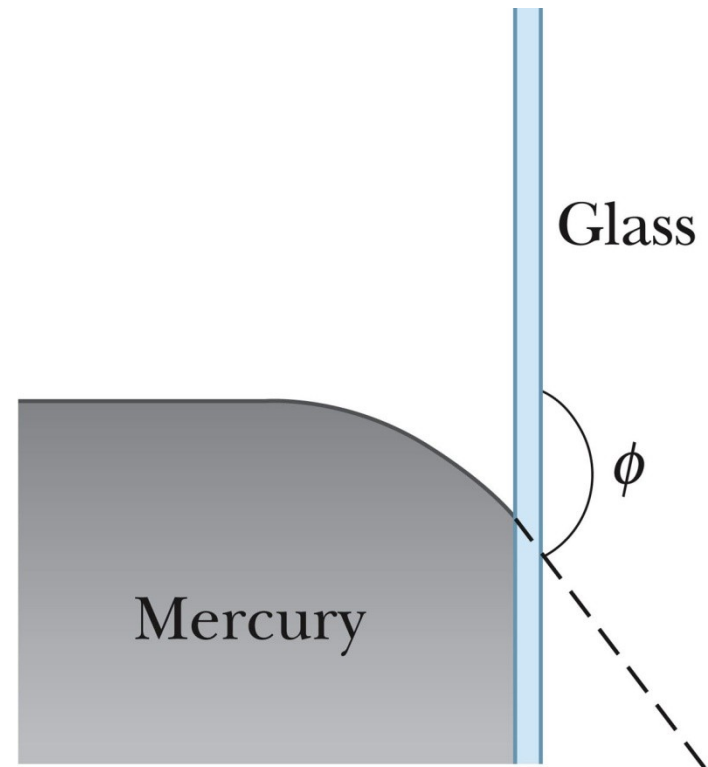
- The adhesive forces are greater than the cohesive forces
- The liquid clings to the walls of the container
- The liquid “wets” the surface



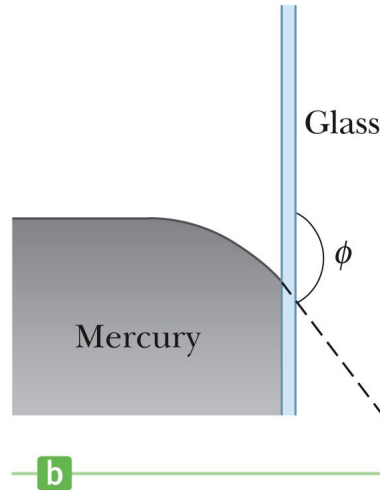
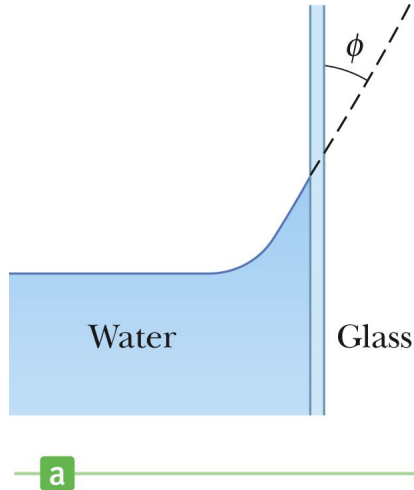
Liquids in Contact with a Solid Surface

– Case 2

- Cohesive forces are greater than the adhesive forces
- The liquid curves downward
- The liquid does not “wet” the surface



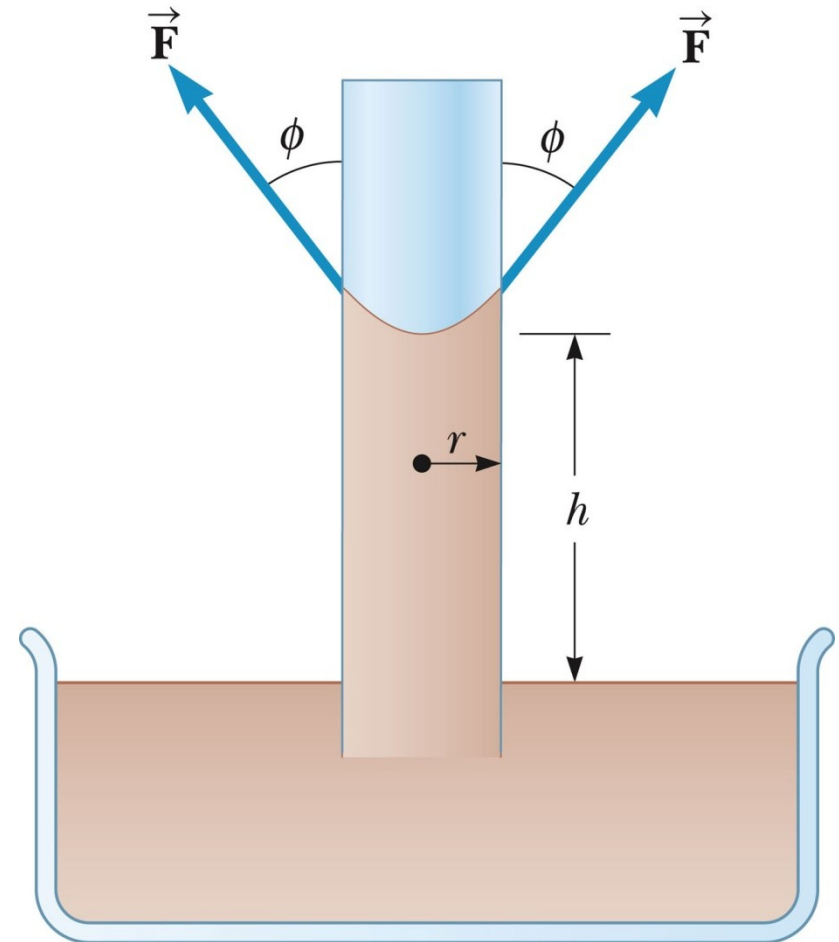
Contact Angle



- The angle, ϕ , between the solid surface and a line drawn tangent to the liquid at the surface is called the **contact angle**
- In a, $\phi > 90^\circ$ and cohesive forces are greater than adhesive forces
- In b, $\phi < 90^\circ$ and adhesive forces are greater than cohesive forces

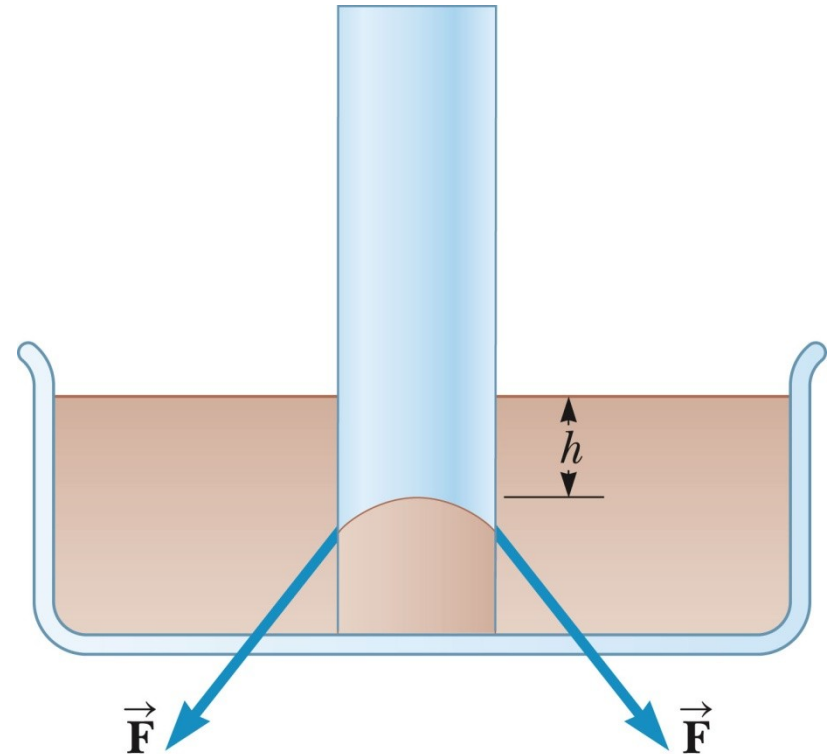
Capillary Action

- Capillary action is the result of surface tension and adhesive forces
- The liquid rises in the tube when adhesive forces are greater than cohesive forces
- At the point of contact between the liquid and the solid, the upward forces are as shown in the diagram



Capillary Action, cont.

- Here, the cohesive forces are greater than the adhesive forces
- The level of the fluid in the tube will be below the surface of the surrounding fluid



Capillary Action, final.

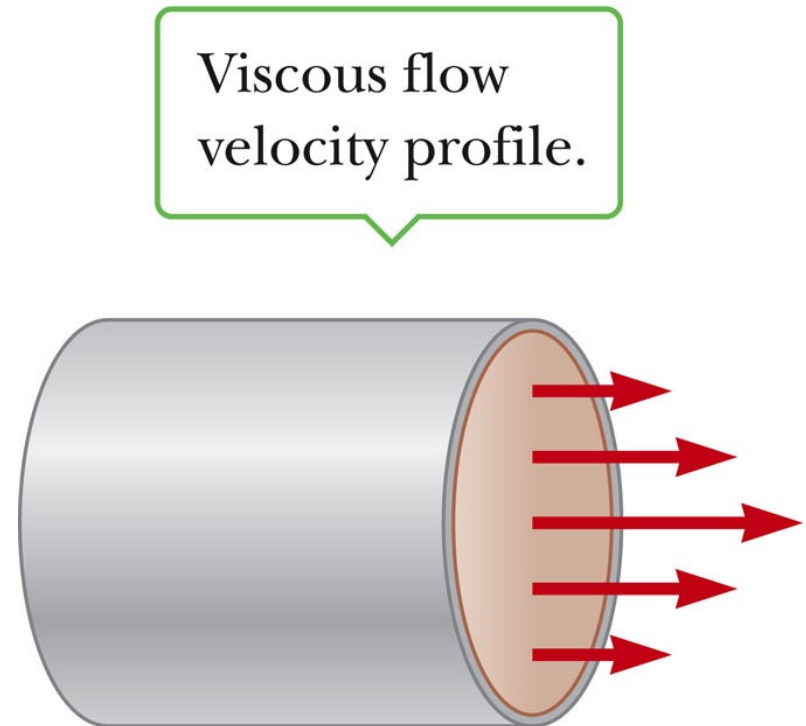
- The height to which the fluid is drawn into the tube is given by:

$$h = \frac{2\gamma}{\rho g r} \cos \varphi$$

- h will also be the distance to the depressed surface

Viscous Fluid Flow

- Viscosity refers to friction between the layers
- Layers in a viscous fluid have different velocities
- The velocity is greatest at the center
- Cohesive forces between the fluid and the walls slow down the fluid on the outside

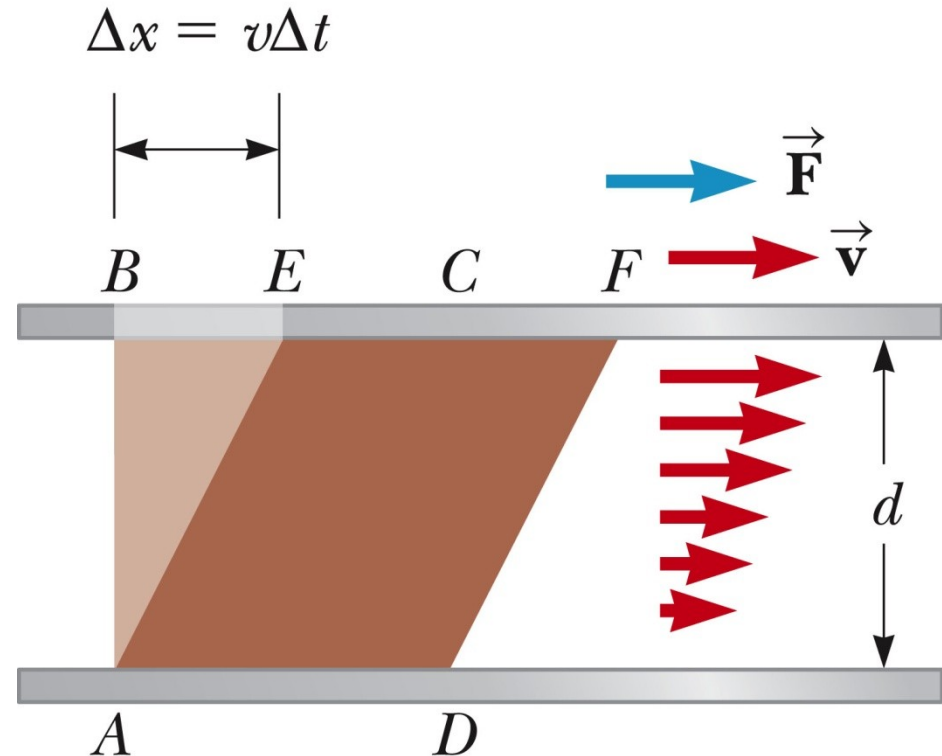


Coefficient of Viscosity

- Assume a fluid between two solid surfaces
- A force is required to move the upper surface

$$F = \eta \frac{Av}{d}$$

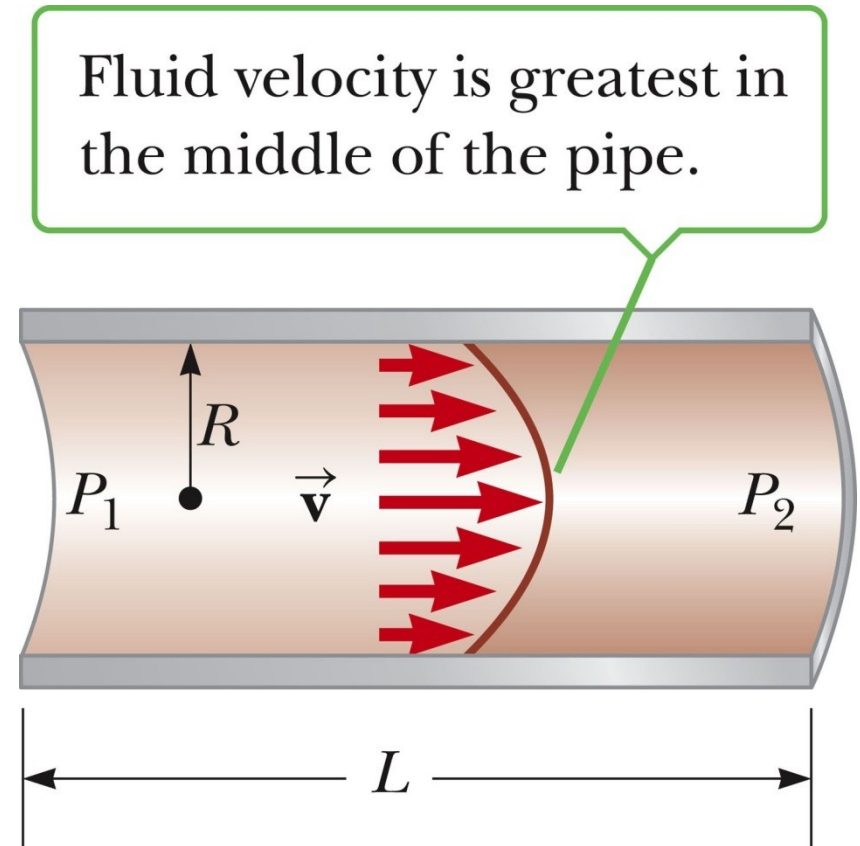
- η is the coefficient of viscosity
- SI units: $\text{N} \cdot \text{s}/\text{m}^2$
- cgs units are Poise
 - 1 Poise = $0.1 \text{ N} \cdot \text{s}/\text{m}^2$



Poiseuille's Law

- Gives the *rate of flow* of a fluid in a tube with pressure differences

$$\text{Rate of flow} = \frac{\Delta V}{\Delta t} = \frac{\pi R^4 (P_1 - P_2)}{8\eta L}$$



Reynold's Number

- At sufficiently high velocity, a fluid flow can change from streamline to turbulent flow
 - The onset of turbulence can be found by a factor called the Reynold's Number, RN

$$RN = \frac{\rho v d}{\eta}$$

- If $RN = 2000$ or below, flow is streamline
- If $2000 < RN < 3000$, the flow is unstable
- If $RN = 3000$ or above, the flow is turbulent

Transport Phenomena

- Movement of a fluid may be due to differences in concentration
 - As opposed to movement due to a pressure difference
 - Concentration is the number of molecules per unit volume
- The fluid will flow from an area of high concentration to an area of low concentration
- The processes are called *diffusion* and *osmosis*

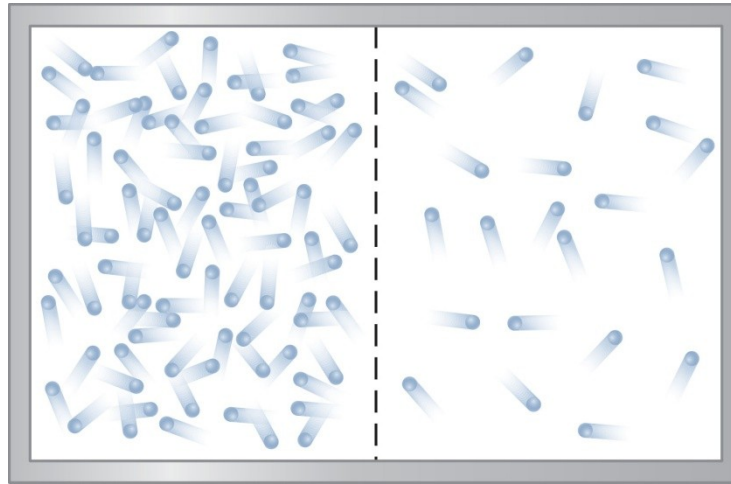
Diffusion and Fick's Law

- Molecules move from a region of high concentration to a region of lower concentration
- Basic equation for diffusion is given by Fick's Law

$$\text{Diffusion rate} = \frac{\text{Mass}}{\text{time}} = DA \left(\frac{C_2 - C_1}{L} \right)$$

- D is the diffusion coefficient

Diffusion



- Concentration on the left is higher than on the right of the imaginary barrier
- Many of the molecules on the left can pass to the right, but few can pass from right to left
- There is a net movement from the higher concentration to the lower concentration

Osmosis

- Osmosis is the movement of water from a region where its concentration is high, across a selectively permeable membrane, into a region where its concentration is lower
 - A *selectively permeable membrane* is one that allows passage of some molecules, but not others

Motion Through a Viscous Medium

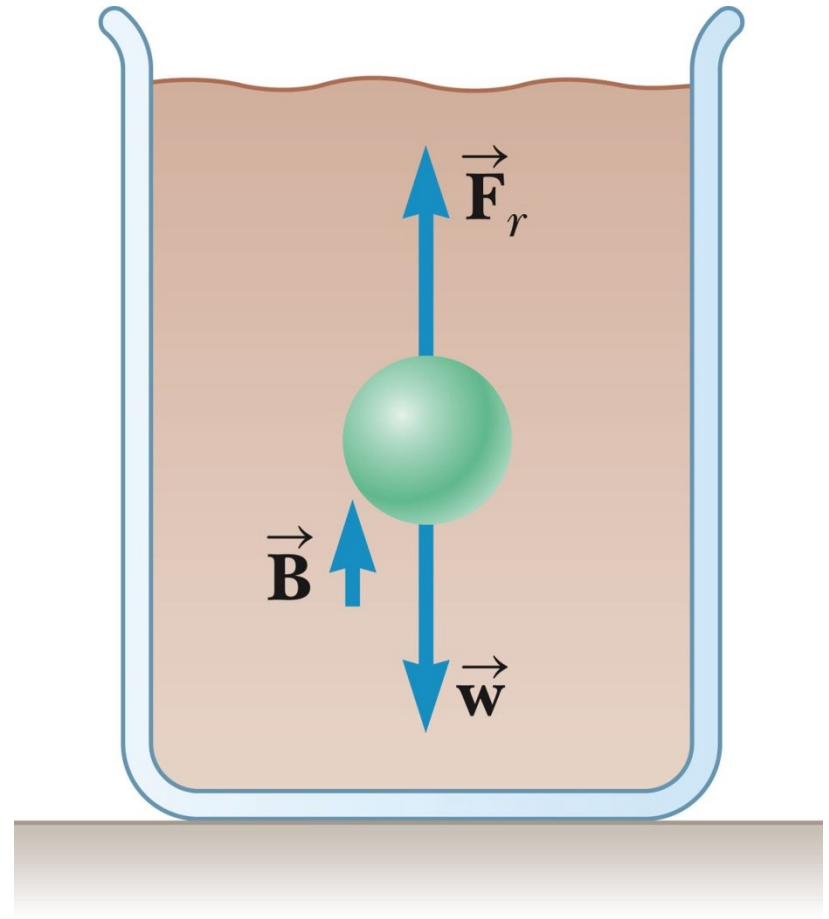
- When an object falls through a fluid, a viscous drag acts on it
 - The force of resistance depends on the shape and velocity of the falling object
- The resistive force on a small, spherical object of radius r falling through a viscous fluid is given by *Stoke's Law*:

$$F_r = 6\pi\eta r v$$

Terminal Velocity

- As the object falls, three forces act on the object
- As its speed increases, so does the resistive force
- At a particular speed, called the terminal speed, the net force is zero

$$v_t = \frac{2r^2g}{9\eta}(\rho - \rho_f)$$



Terminal Velocity, General

- Stokes' Law will not work if the object is not spherical
- Assume the resistive force has a magnitude given by $F_r = k v$
 - k is a coefficient to be determined experimentally
- The terminal velocity will become

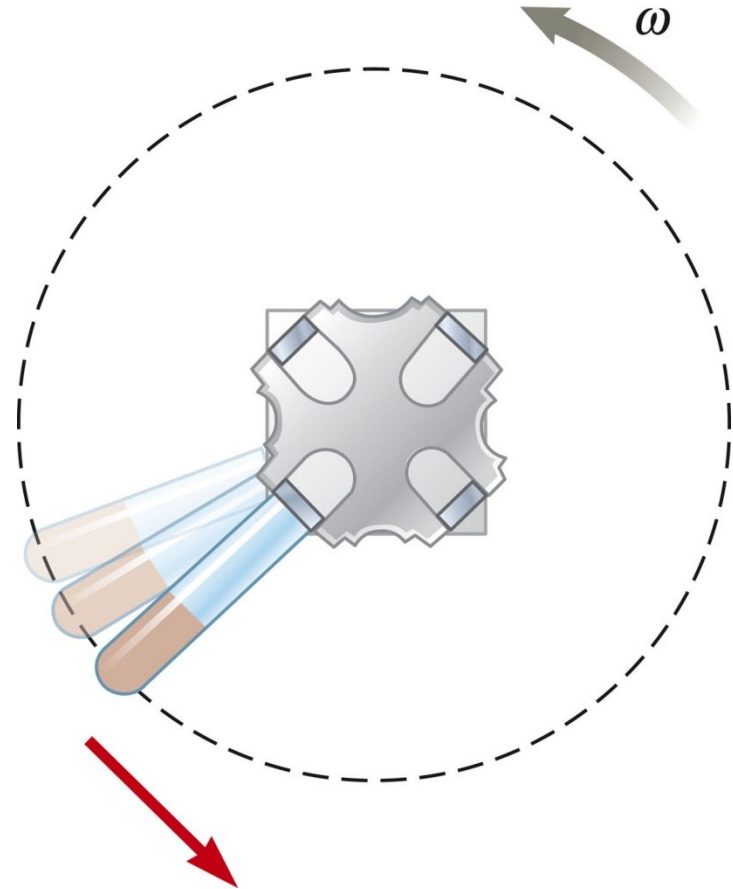
$$v_t = \frac{mg}{k} \left(1 - \frac{\rho_f}{\rho} \right)$$

Sedimentation Rate

- The speed at which materials fall through a fluid is called the **sedimentation rate**
 - It is important in clinical analysis
- The rate can be increased by increasing the effective value of g
 - This can be done in a *centrifuge*

Centrifuge

- High angular speeds give the particles a large radial acceleration
 - Much greater than g
 - In the equation, g is replaced with $\omega^2 r$



Centrifuge, cont

- The particles' terminal velocity will become

$$v_t = \frac{m\omega^2 r}{k} \left(1 - \frac{\rho_f}{\rho} \right)$$

- The particles with greatest mass will have the greatest terminal velocity
- The most massive particles will settle out on the bottom of the test tube first